Project Portfolio Budgeting as a Markov Decision Process: Environment Design, Baseline Benchmarks, and Optimality Gap Analysis

Mahriar Gharaghani*
Seyed Farid Ghannadpour

Department of Industrial Engineering, Iran University of Science and Technology
Tehran, Iran

Abstract

Allocating a shared capital reserve across a portfolio of simultaneously executing contracted projects is a high-frequency, high-stakes decision for engineering contractors. Classical approaches either assume full foresight or collapse the portfolio to a single project, leaving practitioners with no principled tool for the sequential, multi-project setting under real-world uncertainty. This paper closes that gap by delivering a complete, verified simulation environment that makes the problem tractable for data-driven methods.

We formalise the problem as a Constrained Markov Decision Process in which a portfolio manager allocates budget fractions across N concurrent projects at each discrete period. The state encodes Earned Value Management indicators, S-curve planned progress, milestone payment lookaheads, and a per-project breach-tolerance counter. The transition dynamics reproduce the full FIDIC contract cash-flow cycle: advance payment and recovery, progress-weighted milestone gross payments, retention withholding and release, interest charges on treasury draws, and a two-condition termination gate. The reward signal is the γ-discounted portfolio net cash flow, accumulating to an episode NPV that serves as the primary performance criterion.

The environment is released as a configurable, Gymnasium-compatible package with a SQLite audit ledger that records every allocation, cash-flow event, and breach flag — eliminating the black-box problem inherent to learned policies. To establish a performance baseline and motivate the need for learned agents, we run three deterministic greedy heuristics (Reach-Plan, Catchup-T, and Catchup-Next-T) and a manually computed full-information optimum across four benchmark configurations of increasing portfolio complexity. The results show a systematic and widening gap between the greedy baselines and the optimal solution as portfolio complexity grows, confirming that standard rule-based allocation is insufficient and that a well-defined improvement target exists for reinforcement learning agents. Training such agents and evaluating them against a formal Linear Programming bound are identified as the natural next steps, addressed in subsequent work.

Keywords: project portfolio management; budget allocation under performance uncertainty; constrained Markov decision process; reinforcement learning; FIDIC contracts; earned value management; Gymnasium environment; greedy baseline; optimality gap.

*Corresponding author: mahriar_gharaghani@ind.iust.ac.ir

1 Introduction

1.1 Motivation and Problem Context

Large infrastructure and engineering projects routinely overrun their budgets and miss their schedules (Flyvbjerg et al., 2003; Merrow, 2011). When a portfolio of such projects executes simultaneously under a shared capital commitment, the risks compound: a performance shortfall on one project propagates through the shared budget to all others, and the decision of how much to allocate to each project at each period can determine whether the portfolio succeeds or fails.

Despite this, the formal treatment of portfolio-level budget allocation as a sequential decision problem is absent from the project portfolio management literature. Research has concentrated on the *selection* stage — which projects to include — while the *operational governance* stage — how to manage a committed portfolio through to completion — has received comparatively little analytical treatment (Archer and Ghasemzadeh, 1999; Cooper et al., 2001; Turner, 2014). The present work fills this gap.

We consider a portfolio of n projects, each governed by a bilateral contract consistent with FIDIC forms (Fédération Internationale des Ingénieurs-Conseils, 2017), executing concurrently under a relaxed capital commitment. A portfolio manager exercises one recurring decision at each discrete planning period: how much budget to allocate to each active project. The formal problem is to find the allocation policy that maximises discounted net cash flow across the portfolio horizon while respecting contract mechanics.

1.2 The Project Management Decision Hierarchy

The present contribution operates at the portfolio governance layer of the three-tier project management hierarchy (Project Management Institute, 2017). Portfolio selection — which projects to commit to — is treated as an exogenous input, consistent with the two-stage governance model of Turner (2014). At the governance layer, the portfolio manager does not control project execution directly but allocates budget to each project and observes earned value outcomes. Termination is not a deliberate agent action but an endogenous environment mechanic triggered when a project's forecast performance trajectory breaches two conditions simultaneously for a project-specific number of consecutive periods.

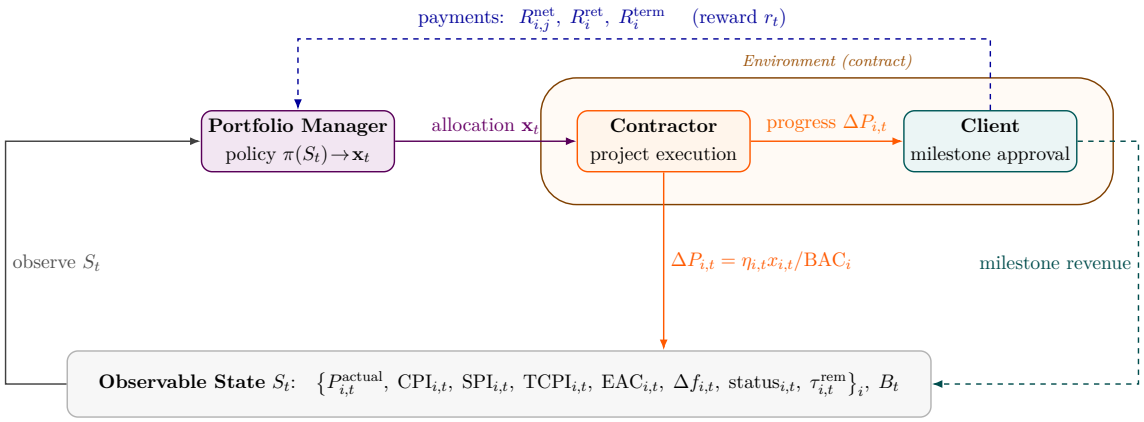


Figure 1: Portfolio governance as a reinforcement learning feedback loop.

1.3 Contract Mechanics and Cash-Flow Structure

The financial interaction between the portfolio manager and each contractor is governed by FIDIC contract forms (Fédération Internationale des Ingénieurs-Conseils, 2017). Three mechanisms determine the cash-flow structure.

Milestone-based payment. Payment is contingent on certified progress thresholds. The gross payment at milestone j is a contractually fixed fraction $\phi_{i,j}$ of the contract price $CP_i = (1 + \pi_i) \cdot \mathrm{BAC}_i$. Because payment is contingent on certified progress, the timing of cash inflows depends on the contractor's actual performance and the portfolio manager's allocation decisions. Cash-flow uncertainty in the present model arises from a single source: the stochastic budget-to-progress efficiency $\eta_{i,t}$, which governs how much physical progress a given budget allocation yields in each period. Uncertain productivity induces uncertain milestone timing, which in turn induces uncertain payment arrival — without requiring any separate model of payment delay. This formulation is sufficient to justify the uncertainty framing of the title and admits a cleaner identification of the allocation problem structure than models with multiple independent uncertainty sources.

Advance payment and progressive recovery. At project initiation, an advance payment $A_i = \alpha_i \cdot CP_i$ is disbursed to the contractor. It is recovered through percentage deductions from subsequent interim payment certificates, with commencement threshold ψ_i and deduction rate δ_i^{rec} specified as project-specific contract parameters (Fédération Internationale des Ingénieurs-Conseils, 2017).

Retention. A fraction ρ_i of each milestone payment is withheld as a performance bond. The accumulated retention $R_i^{\mathrm{ret}} = \rho_i \cdot CP_i$ is released in full upon project completion, creating a structural completion incentive embedded in the payment mechanics.

Earned value management as a performance signal. The EVM framework (Fleming and Koppelman, 2010) provides the diagnostic vocabulary for monitoring contractor performance. The Cost Performance Index $\mathrm{CPI}_{i,t} = \mathrm{BCWP}_{i,t}/\mathrm{ACWP}_{i,t}$ and Schedule Performance Index $\mathrm{SPI}_{i,t} = \mathrm{BCWP}_{i,t}/\mathrm{BCWS}_{i,t}$ are observable state variables. The Estimate at Completion $\mathrm{EAC}_{i,t} = \mathrm{ACWP}_{i,t} + (\mathrm{BAC}_i - \mathrm{BCWP}_{i,t})/\mathrm{CPI}_{i,t}$ projects total cost at the current performance rate and feeds one of the two termination conditions.

Termination. Termination is an endogenous environment mechanic governed by three breach conditions, any one of which decrements a project-specific cure period counter $\tau_{i,t}^{\mathrm{rem}}$ by one: zero allocation to an active project in a given period; simultaneous breach of the schedule tolerance Ω_i by the forecast finish deviation $\Delta f_{i,t}$ and of the cost cap $(1 + \mu_i) \cdot \mathrm{BAC}_i$ by $\mathrm{EAC}_{i,t}$; and plan deviation exceeding the contractual threshold δ^{plan}. Multiple simultaneous breaches decrement the counter by exactly one. The counter resets to τ_i^{tol} only when all three conditions are clear. When the counter reaches zero, or at the hard deadline $f_i + \Omega_i$ if the project is still active, termination executes and the settlement payment $R_i^{\mathrm{term}} = P_{i,t}^{\mathrm{actual}} \cdot CP_i - (A_i + \sum_{j \in \mathcal{M}_i^{\mathrm{cert}}} R_{i,j}^{\mathrm{net}})$ enters the portfolio cash balance.

1.4 The Three-Level Modelling Arc

Level 1: Deterministic full-foresight MILP. When $\eta_{i,t}$ is replaced by a fixed known value $\bar{\eta}_{i,t}$, the progress recursion becomes a deterministic linear map and the problem

collapses to a Mixed-Integer Linear Program. Its optimal value Z^*_{L1} is a provable upper bound on any causal policy operating under uncertainty.

Level 2: Stochastic extension and intractability. When $\eta_{i,t}$ is a random variable, the MILP becomes a multi-stage stochastic program. The scenario tree grows as K^{nH} leaf nodes, rendering exact solution intractable at realistic portfolio scales.

Level 3: Constrained MDP and RL agent. The CMDP reformulation (Altman, 1999) provides a tractable alternative. The policy is optimised through repeated interaction with the environment using sampled trajectories rather than exhaustive scenario enumeration. The CMDP is implemented as a Gymnasium-compatible environment (Towers et al., 2024) and a PPO agent (Schulman et al., 2017) is trained and evaluated against the Level 1 oracle and rule-based baselines.

1.5 Contributions and Paper Organisation

The primary contribution is a configurable Gymnasium-compatible CMDP environment encoding FIDIC contract mechanics, stochastic budget-to-progress efficiency, EVM-based performance signals, and an endogenous termination mechanic with a project-specific cure period counter. The PPO agent trained on this environment constitutes the secondary contribution, evaluated empirically against the Level 1 MILP oracle and rule-based baselines.

Section 2 reviews four literature streams: project management foundations, portfolio selection and governance, PPM budgeting and cash-flow modelling, and reinforcement learning for sequential resource allocation. Section 3 develops the three-level formulation formally. Section **??** presents training protocol and empirical evaluation. Section **??** concludes with implications and future directions.

2 Literature Review

2.1 Overview and Review Strategy

This section maps the intellectual lineage of the present work across four adjacent knowledge domains: project management (PM) foundations, project portfolio management (PPM) and selection, PPM budgeting and cash-flow modelling, and reinforcement learning (RL) applied to sequential resource-allocation problems. Together these streams define both the theoretical foundation and the gap that motivates the proposed framework. The review follows the PRISMA 2020 reporting guidelines (Page et al., 2021) to ensure transparency and reproducibility in source identification, screening, and inclusion decisions.

2.1.1 Search and Screening Protocol

Literature was retrieved from Google Scholar, Semantic Scholar, and Scopus using a progressively narrowing search strategy. Queries began at the broadest relevant domain (*project management*) and were successively refined through *project portfolio management*, *portfolio budgeting and cash-flow optimisation*, *earned value management under uncertainty*, and *reinforcement learning for project scheduling and resource allocation*. Citation-network traversal — backward and forward snowballing from high-citation anchor papers — was used to recover seminal works not surfaced by keyword search.

Records were screened in three sequential stages consistent with the PRISMA 2020 flow diagram (Appendix A). **Stage 1** (identification) assessed machine readability and bibliographic completeness. **Stage 2** (relevance filtering) applied cosine-similarity ranking against the research hypothesis using sentence-transformer embeddings, retaining sources above a calibrated relevance threshold. **Stage 3** (structured extraction) classified each retained paper by contribution type, key variables, modelling paradigm, and empirical basis. The PRISMA flow diagram quantifying records at each stage is provided in Appendix A.

2.2 Project Management: Foundational Concepts

The academic study of project management draws on schedule control (CPM/PERT), earned value management (EVM), and risk modelling as its core quantitative pillars (Fleming and Koppelman, 2010; Project Management Institute, 2021). EVM introduced the Cost Performance Index (CPI) and Schedule Performance Index (SPI) as real-time diagnostic signals relating budgeted cost of work performed to actual expenditure and planned schedule — providing the performance vocabulary that the present model inherits directly. The present model adopts the Earned Schedule correction to SPI (Lipke, 2003), replacing the classical ratio BCWP/BCWS with $\text{SPI}(t) = \text{ES}/\text{AT}$, which continues to degrade past the planned finish date rather than freezing when BCWS reaches BAC — a known limitation of classical SPI in post-planned-finish recovery periods (Lipke, 2003; **?**). The To-Complete Performance Index (TCPI) is also included as a forward-looking complement to CPI, measuring the cost efficiency required on remaining work to finish within budget. Values of $\text{CPI} < 1$ signal cost overrun; $\text{SPI} < 1$ signals schedule lag. Stochastic extensions of EVM (Acebes et al., 2014; Batselier and Vanhoucke, 2017) and Monte Carlo simulation frameworks (Kwak and Ingall, 2007) have modelled performance uncertainty at the project level, establishing that CPI and SPI follow empirically characterisable distributions over project life. Empirical evidence on contractor performance variability from EVM datasets (Acebes et al., 2014; Batselier and Vanhoucke, 2017) provides the calibration basis for the stochastic efficiency parameter $\eta_{i,t}$ that parameterises the stochastic program at Level 2 and the CMDP environment at Level 3.

S-curve analysis, formalised by Kenley and Wilson (2009) for construction cash flows, established that budget consumption follows characteristic sigmoid profiles whose shape parameters are project-type- and contract-form-dependent. The regularized incomplete Beta function provides a flexible single-family parameterisation of this profile (Barraza et al., 2004), and is adopted directly in the planned expenditure model of the present formulation.

Large-scale empirical evidence on project performance is provided by Merrow (2011) through the Independent Project Analysis database, and by Flyvbjerg et al. (2003); Flyvbjerg (2017) through infrastructure cost-overrun datasets spanning decades and geographies. These datasets inform both the calibration of uncertainty distributions in the present model and the benchmarks against which synthetic environment parameters are validated. Complementary empirical benchmarks are drawn from AACE International and Construction Industry Institute cost-engineering databases (Christensen and Heise, 1993).

Contract governance structures — particularly FIDIC forms (Red, Yellow, and Silver Books) — prescribe the financial interaction between owner and contractor through milestone payment schedules, advance payment and recovery mechanisms, and retention clauses (Fédération Internationale des Ingénieurs-Conseils, 2017). The Estimate at Completion $\text{EAC}_{i,t} = \text{ACWP}_{i,t} + (\text{BAC}_i - \text{BCWP}_{i,t})/\text{CPI}_{i,t}$ extends CPI into a forward-looking cost trajectory signal; Christensen and Heise (1993) established across 155 contracts that

this formulation produces the most accurate and stable cost-at-completion forecasts, with reliability increasing beyond the 20% completion mark (Lipke, 2003). The present model adopts EAC as the second termination condition signal, directly parallel to the forecast finish deviation used in Condition 1. Under FIDIC Sub-Clause 14.2, the advance payment is an interest-free mobilisation loan recovered through percentage deductions from interim payment certificates: if not otherwise specified in the Appendix to Tender, deductions commence once cumulative certified interim payments exceed 10% of the Accepted Contract Amount and are applied at a rate of 25% of each subsequent payment certificate until the advance is fully recovered (Fédération Internationale des Ingénieurs-Conseils, 2017). The present model parameterises this mechanism with a project-specific commencement threshold ψ_i and deduction rate δ_i^{rec}, accommodating the full range of particular conditions encountered in practice. These mechanisms translate directly into the cash inflow and outflow dynamics formulated in Section **??**.

2.3 Project Portfolio Management: Selection and Governance

Project portfolio management formalises the problem of allocating an organisation's finite resources across a competing set of candidate projects to maximise strategic value (Archer and Ghasemzadeh, 1999; Cooper et al., 2001). The literature distinguishes two nested decision layers: *portfolio selection* — the binary or mixed-integer optimisation of which projects to include — and *portfolio governance* — the ongoing operational oversight of approved, active projects.

Portfolio selection models have been extensively developed. Markowitz-style mean–variance frameworks adapted from financial portfolio theory (Markowitz, 1952) have been applied to project baskets to trade off expected NPV against variance (Eilat et al., 2006). Multi-criteria decision analysis extensions incorporate strategic alignment, risk ratings, and resource interdependencies (Martinsuo and Lehtonen, 2007; Killen et al., 2008). Metaheuristic approaches — genetic algorithms, particle swarm optimisation — address large combinatorial instances (Zhang et al., 2020). Resource-constrained portfolio selection under capital rationing has been studied by Kavadias and Loch (2003), whose no-borrowing constraint is adopted directly in the present formulation.

The present work *explicitly adopts the output of portfolio selection as an exogenous input*. The selection problem is treated as solved upstream; the focus is entirely on the operational governance layer that receives a committed portfolio and must manage it through execution. This decomposition is consistent with the two-stage governance model of Turner (2014) and the PMI Standard for Portfolio Management (Project Management Institute, 2017), which separates strategic selection from operational execution oversight. In doing so, this work builds directly on the selection literature while advancing the frontier into the governance domain that has received substantially less formal analytical treatment.

2.4 PPM Budgeting, Cash-Flow Modelling, and Uncertainty

Within portfolio governance, budget allocation across active projects over time represents a high-frequency, high-stakes operational decision. Despite its practical centrality, this specific problem has received limited formal treatment relative to portfolio selection. Existing work addresses project-level cash-flow modelling (Kenley and Wilson, 2009; Park and Peña-Mora, 2004), EVM-based forecasting of individual project cost-at-completion (Kim and Ballard, 2002; Acebes et al., 2014), and multi-project resource levelling (Lova et al., 2000; Van Peteghem and Vanhoucke, 2010). However, these contributions treat

budgets as deterministic constraints rather than as dynamic decision variables subject to portfolio-level sequential optimisation.

Cash-flow uncertainty in the present model arises from a single source: the budget-to-progress efficiency $\eta_{i,t}$, which governs how much physical progress a given budget allocation yields in each period (Acebes et al., 2014; Batselier and Vanhoucke, 2017). Uncertain productivity determines whether and when each milestone threshold is crossed, which in turn determines the timing of the contractually fixed payment $R^{\text{net}}_{i,j}$. Payment delays between certification and receipt are not modelled as an independent uncertainty source. This relaxation is defensible: the performance-driven channel is the structurally primary source of cash-flow timing uncertainty in milestone-payment contracts, and isolating it yields a cleaner identification of the allocation problem without loss of the essential uncertainty framing. Other sources of cash-flow uncertainty — scope changes, external market shocks (Doloi, 2012) — are treated as exogenous and outside the scope of the present formulation. Stochastic programming (Birge and Louveaux, 2011) provides the canonical OR framework for optimisation under such uncertainty, and its two-stage variant has been applied to project scheduling under resource uncertainty (Lova et al., 2000). However, multi-stage stochastic programs for portfolio-level sequential budget allocation face scenario-tree explosion as the number of projects and periods increases. The single uncertainty source $\eta_{i,t}$ now branches the scenario tree at every period: if $\eta_{i,t}$ takes K discrete values at each of H periods for each of n projects, the tree produces K^{nH} leaf nodes. The effective planning horizon is itself portfolio-dependent—determined by the latest project finish—compounding the explosion, since H grows with the most delayed project in the committed set. Both effects are formalised in Section 3.5.

The gap in the literature is precise: *no established framework models the portfolio-level budget allocation decision as a sequential stochastic optimisation problem in which a central decision-maker allocates funds across multiple active projects whose cash inflow trajectories are uncertain through stochastic budget-to-progress efficiency, subject to contractual milestone certification windows that bound the speed at which progress converts to cash inflows.* The present work addresses this gap. The objective is pure discounted net cash flow; no deviation penalty enters the objective function. This is a deliberate design choice: the consequences of sustained deviation — termination and loss of all future milestone payments and retention — are sufficient to make deviation undesirable without a separate penalty term, and including one would introduce an arbitrary scalarisation weight with no natural calibration from contract data (Ng et al., 1999). Symmetrically, the discount factor γ encodes not only time preference but the opportunity cost of tied-up capital in a competitive environment where future projects are foregone while current ones remain incomplete (Damodaran, 2012). This interpretation gives γ a concrete calibration anchor beyond abstract preference — it can be estimated from the firm's observed pipeline redeployment rates or from prevailing cost-of-capital benchmarks. The termination settlement further grounds the model in real asset values: the settlement payment $R^{\text{term}}_i = P^{\text{actual}}_{i,t} \cdot CP_i - (A_i + \sum_{j \in \mathcal{M}^{\text{cert}}_i} R^{\text{net}}_{i,j})$ computes the difference between the contractor's entitlement for work delivered and cumulative inflows already received, which can be positive or negative depending on progress at termination relative to certified payments (Dixit and Pindyck, 1994; Huchzermeier and Loch, 2001), rather than an arbitrarily weighted deviation penalty. A consequence is that the learned policy may produce allocation trajectories that concentrate expenditure around milestone certification thresholds, mirroring documented contractor financial strategy in milestone-payment contracts (Kenley and Wilson, 2009; Kaka and Price, 1993; Arditi and Chotibhongs, 2005).

2.5 Contractor Underperformance, Cure Periods, and Termination

When a contractor fails to meet schedule or cost performance thresholds, international EPC contracts prescribe a structured escalation sequence before termination may be exercised. Under FIDIC Sub-Clause 15.1, the client must issue a formal notice of default and allow a cure period — typically 28 days — during which the contractor is obliged to demonstrate measurable recovery before the client may proceed to termination (Fédération Internationale des Ingénieurs-Conseils, 2017; Ndekugri and Rycroft, 2008). This cure period is not a discretionary grace; it is a contractual obligation whose omission constitutes wrongful termination. Empirical studies of international EPC disputes confirm that failure to observe the cure period is among the most frequently litigated grounds in contractor termination claims (Ndekugri and Rycroft, 2008; Ennis and Burr, 2021).

The behavioural project management literature has studied the conditions under which organisations fail to terminate underperforming projects at the economically rational point. Escalation of commitment (Staw and Ross, 1987) — the tendency to continue investing in failing projects beyond what expected value justifies — has been documented across construction, IT, and infrastructure programmes. Real options theory (Dixit and Pindyck, 1994; Huchzermeier and Loch, 2001) provides the normative counterpart: termination is the exercise of an abandonment option whose value exceeds continuation NPV once performance has deteriorated past a recoverable threshold. Together these streams establish that the decision to persist with or terminate an underperforming project is neither purely contractual nor purely financial — it involves performance signals, contractual obligations, and timing all at once.

The present model integrates all three dimensions into a single environment mechanic. The cure period counter $\tau_{i,t}^{\text{rem}}$ decrements by exactly one whenever any of three breach conditions is active: zero allocation to an active project; simultaneous breach of the contractual schedule tolerance Ω_i by the forecast finish deviation and of the cost cap $(1+\mu_i)\cdot \text{BAC}_i$ by $\text{EAC}_{i,t}$; or plan deviation exceeding a contractual threshold, grounded in the FIDIC engineer's authority to issue a notice when progress falls materially behind programme (Fédération Internationale des Ingénieurs-Conseils, 2017). The schedule–cost condition is forward-looking and symmetric in structure: remaining planned work divided by current performance rate, projected forward to completion (Fleming and Koppelman, 2010; Christensen, 1993). Multiple simultaneous breaches decrement the counter by exactly one. The counter resets to τ_i^{tol} only when all three conditions are fully clear; partial resolution does not reset it. Termination executes when the counter reaches zero, or unconditionally at the hard deadline $f_i + \Omega_i$ if the project remains active. This mechanic is grounded simultaneously in FIDIC contract law (Fédération Internationale des Ingénieurs-Conseils, 2017; Ndekugri and Rycroft, 2008), the real options framing of the abandonment threshold (Dixit and Pindyck, 1994; Huchzermeier and Loch, 2001), the EVM literature's validation of EAC as a stable forward cost signal (Christensen, 1993; Lipke, 2003), and the escalation-of-commitment literature's evidence that observable performance signals must be sustained — not transient — before termination is justified (Staw and Ross, 1987). The tolerance counter is part of the observable portfolio state, giving the RL agent an anticipatory signal that reflects contractual reality without granting it discretionary control over the termination outcome. From the portfolio manager's perspective, this anticipatory signal serves a dual function — it flags projects approaching contractual failure while also signalling the point at which continued allocation yields diminishing returns relative to the settlement value of work already delivered. A rational manager reading a counter approaching zero in a high-opportunity-cost environment would recognise that the real value of future milestone payments is eroding while the settlement value of work in place

is fixed in nominal terms — creating a natural reallocation incentive that the model captures through the interaction of γ and R_i^{term} without requiring explicit inflation modelling (Dixit and Pindyck, 1994; Fédération Internationale des Ingénieurs-Conseils, 2017).

2.6 Reinforcement Learning for Sequential Resource Allocation

Reinforcement learning provides a principled framework for sequential decision-making under uncertainty (Sutton and Barto, 2018). The MDP formalisation — state, action, transition, reward — maps naturally onto the portfolio budget allocation problem, and the constrained MDP extension (Altman, 1999) accommodates the hard budget feasibility constraints that the portfolio manager must satisfy at every period.

RL has been applied to project scheduling (Peng and Huang, 2021; Liu et al., 2023), resource-constrained project scheduling problems (RCPSP) (Tassel et al., 2023), and supply chain inventory management (Giannoccaro and Pontrandolfo, 2002). In each domain, RL demonstrates advantages over classical OR methods when state spaces are large, transition dynamics are stochastic, the problem is non-stationary over the decision horizon, and the horizon length is itself uncertain at the outset of planning. These are precisely the characteristics of the portfolio budget allocation problem at Levels 2 and 3.

Proximal Policy Optimisation (Schulman et al., 2017) is adopted as the solution algorithm for its stability in continuous action spaces, its sample efficiency relative to earlier policy-gradient methods, and its established use in operations management domains. Domain randomisation (Tobin et al., 2017) — randomising environment parameters during training to improve out-of-sample policy robustness — is adopted to ensure the learned policy generalises across the range of project types, contract structures, and market conditions represented in the calibration data. Walk-forward validation (Cerqueira et al., 2020) is used as the empirical evaluation protocol to guard against data leakage given the time-series structure of project cash-flow data.

Direct application of RL to portfolio-level budget allocation remains absent from the literature. The present work provides the first formulation of this problem as a CMDP and the first empirical demonstration that a PPO agent can learn non-trivial allocation policies in a calibrated synthetic environment.

2.7 Synthesis: Positioning the Present Contribution

Table 1 maps representative works across the four literature streams against the key modelling dimensions of the present framework. Three structural gaps emerge.

First, the transition from portfolio *selection* to portfolio *operational governance* has not been modelled as a sequential decision problem. Selection literature assumes away post-selection management; project scheduling literature ignores portfolio interdependencies and the shared capital constraint.

Second, existing stochastic models of project cash-flow uncertainty operate at the individual project level and do not feed uncertainty upward into a portfolio-level decision architecture with a shared budget constraint.

Third, the endogenous mechanics of project failure — the joint modelling of forecast schedule deviation, EAC-based cost overrun trajectory, a contractual cure period, and termination settlement — have not been embedded within a portfolio-level sequential decision framework in which budget allocation is the sole control variable.

These gaps are not isolated oversights; they reflect an incomplete progression along the path toward a deployable portfolio-level decision-support system. Figure 2 situates

this progression as three sequential steps. **Step 1** — domain grounding — is well established by the literature reviewed above: contract mechanics, EVM performance signals, S-curve expenditure profiles, and portfolio selection models collectively provide the inputs and boundary conditions that any operational governance model requires. **Step 2** — tractable problem formulation — is the gap. Bridging it requires a relaxation hierarchy: the governance problem must first be stated precisely under idealised assumptions (a deterministic MILP establishing a performance ceiling), then shown to be intractable under realistic stochastic assumptions (exponential scenario-tree growth), and finally reformulated into a tractable approximation (a CMDP solved by a policy-gradient agent). No existing work completes this hierarchy at the portfolio level. **Step 3** — validated deployment within incumbent project management information systems such as Oracle Primavera P6 — remains future work, contingent on the tractable formulation that Step 2 must first provide.

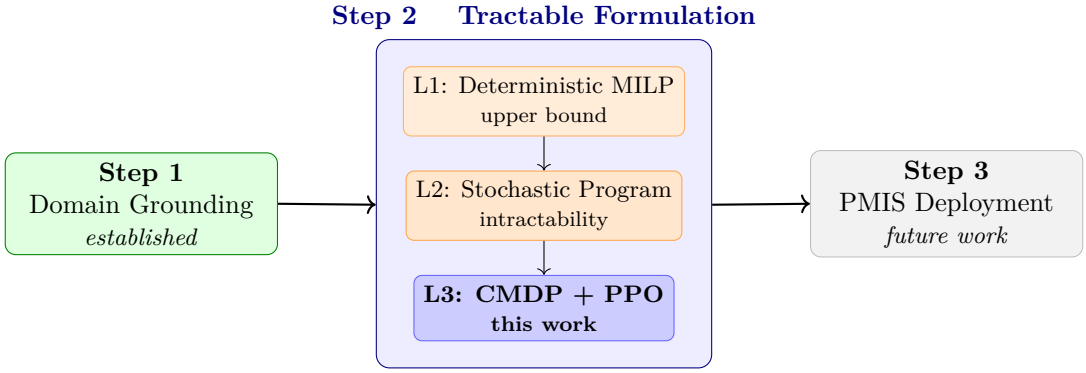


Figure 2: Solution maturity progression toward deployable portfolio-level budget allocation.

The present work addresses all three gaps and completes Step 2 of this progression through the relaxation hierarchy described above and developed formally in Section **??**. The Gymnasium-compatible CMDP environment — with parameterisable contract mechanics, stochastic budget-to-progress efficiency as the single uncertainty source, and an endogenous termination mechanic — constitutes the primary methodological contribution. The PPO agent demonstrates its utility and provides the basis for empirical evaluation against classical OR baselines in Section **??**.

Table 1: Representative literature mapped against modelling dimensions of the present framework. *Decision type*: S = selection, Sc = scheduling, B = budgeting. ✓ = fully addressed; ○ = partially addressed; – = not addressed.

Work	Portfolio level	Decision type	Sequential decisions	Stochastic cash flows	Contract mechanics	Termination settlement
Yamakawa et al. (2019)	✓	S	–	○	–	–
Kosztyán et al. (2022)	✓	S	–	○	–	–
Cooper et al. (2001)	✓	–	–	○	–	–
Touran (2006)	✓	Sc/B	–	○	–	–
Touran (2010)	✓	Sc/B	–	○	–	–
Ghapanchi et al. (2012)	✓	S	–	○	–	–
Pourahmadi et al. (2015)	✓	S	–	○	–	–
El-Sabek and McCabe (2018)	–	B	–	–	✓	–
Baqeri et al. (2019)	✓	S	–	○	–	–
Mohagheghi et al. (2020)	✓	–	–	○	–	–
Beseiso and Kumar (2021)	✓	S	–	○	–	–
Karimi et al. (2021)	✓	S	–	○	–	–
Bahroun et al. (2023)	–	Sc	✓	–	–	–
Ranjbar et al. (2022)	✓	S/Sc	–	–	○	–
Szwarcfiter et al. (2023)	✓	Sc	✓	○	–	–
Asghari et al. (2023)	–	–	✓	○	–	–
Bai et al. (2023)	✓	–	–	○	–	–
Pappert and Kusanke (2023)	✓	S/B	–	–	–	–
Mazhari and Tajik (2024)	✓	S/B	✓	–	–	–
Jiang et al. (2024)	✓	–	✓	○	–	–
Ebnerasoul et al. (2023)	✓	S	–	○	–	–
Noroozi et al. (2016)	✓	B	–	○	–	–
Jiang et al. (2023)	–	B	✓	–	–	–
Shou and Huang (2010)	✓	S/Sc	–	–	–	–
Altman (1999)	–	–	✓	–	–	–
Present work	✓	**B**	✓	✓	✓	✓

3 Problem Formulation

The formulation proceeds across three levels of increasing generality, all sharing the environment primitives defined in this section. **Level 1** (Section 3.4) establishes the deterministic full-foresight MILP whose optimum is a provable upper bound on any causal policy. **Level 2** (Section 3.5) introduces stochastic contractor efficiency and characterises the resulting multi-stage stochastic program, establishing exponential scenario-tree growth. **Level 3** (Section 3.6) reformulates the intractable stochastic program as a Constrained Markov Decision Process, yielding the Gymnasium-compatible environment on which the PPO agent is trained. The three levels form a coherent arc — Level 1 defines the performance ceiling by fixing the sole uncertain primitive ($\eta_{i,t}$) to a known scenario value, Level 2 explains why the resulting stochastic program is unreachable by causal policies, Level 3 provides the tractable path.

3.1 Problem Statement and Scope

The problem is operational, not strategic. A portfolio of n contracted projects is fixed: commitments are signed, start dates are scheduled, and contracts are parameterised consistently with FIDIC Red, Yellow, and Silver Book forms (Fédération Internationale des Ingénieurs-Conseils, 2017). The decision-maker — a portfolio manager, board, or programme director — exercises a single recurring lever at each discrete planning period $t \in \mathcal{T} = \{1, \ldots, H\}$: the budget $x_{i,t} \geq 0$ directed to each active project i. The planning horizon H is not a fixed configuration parameter but is determined endogenously as $H = \max_i f_i - \min_i s_i + 1$, where s_i and f_i are the planned start and finish of project i; the index is shifted so that period $t = 1$ aligns with the earliest project start. All outcomes of interest — milestone achievement, cash inflows, schedule adherence, and project termination — flow through this allocation decision. Portfolio selection is treated as an exogenous upstream input (Archer and Ghasemzadeh, 1999).

Four structural assumptions underlie the entire formulation. (i) *Independent bilateral contracts*: projects interact only through the shared cash balance; no resource or workforce sharing occurs, consistent with standard EPC contracting practice. (ii) *Budget as the sole control*: the contractor's internal resource decisions are absorbed into a stochastic efficiency parameter $\eta_{i,t}$; the portfolio manager observes how much physical progress a budget allocation yields, not how it is achieved. $\eta_{i,t}$ is the sole stochastic primitive: it is unobservable at the time allocation $x_{i,t}$ is committed, and its realisation determines whether and when each milestone threshold is crossed, which in turn determines the timing of cash inflows. The timing of that allocation — specifically how quickly milestones are certified — determines the discounted value of inflows under $\gamma < 1$, making the scheduling of spend, not just its total magnitude, the operative decision variable for a return-maximising portfolio manager (Damodaran, 2012). (iii) *Discrete monthly periods*: $\mathcal{T} = \{1, \ldots, H\}$, where H is determined dynamically from the portfolio (see Section 3.2.1), reflects the standard monthly governance cycle of portfolio management. A relaxed budget regime ($\kappa = B_1 / \sum_i \mathrm{BAC}_i \gg 1$) is adopted throughout, so the liquidity constraint is non-binding and the formulation isolates allocation timing and performance management as the operative decision dimensions. (iv) *FIDIC contract mechanics*: milestone payments, advance payment and recovery, retention, and termination settlement are all calibrated parameters, not fixed constants, accommodating the full range of particular conditions encountered in practice.

3.2 Environment Definition

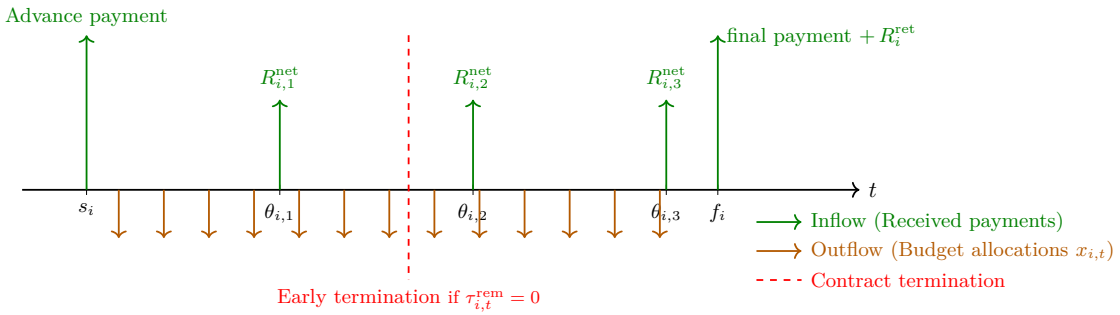


Figure 3: Cash-flow timeline for project i under FIDIC contract mechanics.

3.2.1 Sets, Indices, and the Decision Variable

$\mathcal{P}$	Projects $\{1, \ldots, n\}$, indexed by i; all committed at horizon start.
$\mathcal{T}$	Planning periods $\{1, \ldots, H\}$, indexed by t (calendar months), where $H = \max_{i \in \mathcal{P}} f_i - \min_{i \in \mathcal{P}} s_i + 1$ is derived endogenously from the committed portfolio; period $t = 1$ corresponds to the calendar month of the earliest project start $\min_i s_i$.
$\mathcal{M}_i$	Ordered milestones $\{1, \ldots, M_i\}$ for project i, indexed by j, with progress thresholds $\theta_{i,1} < \cdots < \theta_{i,M_i} = 1$.
$\mathcal{A}_i^{\text{plan}}$	Planned execution window $\{s_i, \ldots, f_i\}$; anchors S-curve computations.
$\mathcal{P}_t^{\text{active}}$	Projects active at period t; fixed at Level 1, endogenous at Levels 2–3.
$\mathcal{M}_i^{\text{cert}}(t)$	Milestones of project i certified by period t: $\{j : \theta_{i,j} \leq P_{i,t}^{\text{actual}}$ and $t \geq e_{i,j}\}$, where $e_{i,j} \in \mathcal{T}$ is the contractually specified earliest certification period for milestone j.
$x_{i,t} \geq 0$	**Decision variable**: budget allocated to project i at period t. The period-t action vector is $\mathbf{x}_t = (x_{i,t})_{i \in \mathcal{P}_t^{\text{active}}}$.
$B_t \geq 0$	Portfolio cash balance at the start of period t; B_1 is given.

3.2.2 Progress and Expenditure Dynamics

Budget $x_{i,t}$ converts to physical progress through the efficiency scalar $\eta_{i,t}$, which absorbs all sources of conversion uncertainty (contractor productivity, supply chain, weather):

$$P_{i,t}^{\text{actual}} = P_{i,t-1}^{\text{actual}} + \eta_{i,t} \cdot \frac{x_{i,t}}{\text{BAC}_i}, \qquad \text{ACWP}_{i,t} = \text{ACWP}_{i,t-1} + x_{i,t}, \tag{1}$$

with $P_{i,0}^{\text{actual}} = \text{ACWP}_{i,0} = 0$. Here BAC_i is the Budget at Completion — the total planned contractor expenditure for project i, which serves as the EVM cost basis and the normalisation denominator for all progress and performance calculations. At Level 1, $\eta_{i,t}$ is replaced by a known deterministic value $\bar{\eta}_{i,t}$; at Levels 2–3 it is the sole stochastic primitive, realised only after $x_{i,t}$ is committed.

The planned cumulative expenditure fraction follows a normalised S-curve (Kenley and Wilson, 2009; Barraza et al., 2004) parameterised by shape parameters (a_i, b_i):

$$P_{i,t}^{\text{plan}} = I_{\tau_{i,t}^c}(a_i, b_i), \qquad \tau_{i,t} = \frac{t - s_i + 1}{D_i^{\text{plan}}}, \quad \tau_{i,t}^c = \min\{1, \max\{0, \tau_{i,t}\}\}, \tag{2}$$

where $I_\tau(\cdot,\cdot)$ is the regularized incomplete Beta function and $D_i^{\text{plan}} = f_i - s_i + 1$ is the planned duration. This profile is a fixed parameter derived at commitment; it is never a decision variable.

3.2.3 Contractual Cash-Flow Mechanics

Payment inflows follow FIDIC mechanics (Fédération Internationale des Ingénieurs-Conseils, 2017). At project initiation ($t = s_i$), the portfolio manager receives the advance payment $A_i = \alpha_i \cdot CP_i$, where α_i is the advance ratio and $CP_i = \text{BAC}_i(1 + \pi_i)$ is the contract price with profit margin π_i. The contract price is the sole payment accumulation baseline: all milestone revenues, advance payment, retention, and final settlement sum to CP_i. All EVM performance signals and progress calculations are normalised against BAC_i.

Milestone j is certified at the first period t at which $P_{i,t}^{\text{actual}} \geq \theta_{i,j}$ *and* $t \geq e_{i,j}$, where $e_{i,j}$ is the contractually specified earliest certification period. Under FIDIC Sub-Clause 14.3, interim payment applications may not be submitted before the intervals stated in the Appendix to Tender (Fédération Internationale des Ingénieurs-Conseils, 2017); $e_{i,j}$ parameterises this constraint directly, preventing a portfolio manager from accelerating cash inflows beyond the contractual payment schedule regardless of how quickly physical progress is achieved. The net payment received, after retention withholding at rate ρ_i and advance recovery deduction at rate δ_i^{rec} (commencing once cumulative certified payments exceed threshold ψ_i), is:

$$R_{i,j}^{\text{net}} = \phi_{i,j} \cdot CP_i(1 - \rho_i) - \delta_i^{\text{rec}} \cdot \phi_{i,j} \cdot CP_i \cdot \mathbf{1}\left\{\textstyle\sum_{k \leq j} \phi_{i,k} \geq \psi_i\right\}, \tag{3}$$

where $\phi_{i,j}$ is the contractual weight of milestone j with $\sum_j \phi_{i,j} = 1$. This parameterisation is consistent with FIDIC Sub-Clause 14.2, under which advance recovery deductions commence once cumulative certified interim payments exceed a specified fraction of the contract price (Fédération Internationale des Ingénieurs-Conseils, 2017). Upon project completion ($P_{i,t}^{\text{actual}} = 1$), the accumulated retention $R_i^{\text{ret}} = \rho_i \cdot CP_i$ is released together with the final milestone payment. The payment identity $A_i + \sum_j R_{i,j}^{\text{net}} + R_i^{\text{ret}} = CP_i$ holds exactly.

Milestone payments are received in the same period they are certified (Relaxation B). The aggregate period inflow is:

$$V_t = \sum_{i,j} R_{i,j}^{\text{net}} \cdot \mathbf{1}\{\tau_{i,j}^{\text{cert}} = t\} + \sum_i A_i \cdot \mathbf{1}\{t = s_i\} + \sum_{i \in \mathcal{P}_t^{\text{term}}} R_i^{\text{term}}, \tag{4}$$

where $\mathcal{P}_t^{\text{term}}$ collects projects newly terminated at t, with settlement R_i^{term} defined in Section 3.2.5 below.

3.2.4 Performance Monitoring: EVM Signals

The Earned Value Management framework (Fleming and Koppelman, 2010) provides the diagnostic vocabulary used to monitor contractor performance and to drive the termination mechanic. The Earned Value at period t is $\text{EV}_{i,t} = P_{i,t}^{\text{actual}} \cdot \text{BAC}_i$ (equivalently, the

Budgeted Cost of Work Performed, BCWP), expressed in cost units against the BAC_i baseline. Three derived indices are computed each period:

$$\text{SPI}_{i,t} = \frac{\text{ES}_{i,t}}{t - s_i + 1}, \tag{5}$$

$$\text{CPI}_{i,t} = \frac{\text{EV}_{i,t}}{\text{ACWP}_{i,t}} = \frac{P_{i,t}^{\text{actual}} \cdot \text{BAC}_i}{\text{ACWP}_{i,t}}, \tag{6}$$

$$\text{EAC}_{i,t} = \text{ACWP}_{i,t} + \frac{\text{BAC}_i - \text{EV}_{i,t}}{\text{CPI}_{i,t}}. \tag{7}$$

The Earned Schedule $\text{ES}_{i,t}$ is the planned time at which the s-curve (2) would have reached the current actual progress level $P_{i,t}^{\text{actual}}$; formally it is the value $\tau^* \in [0, D_i^{\text{plan}}]$ satisfying $I_{\tau^*/D_i^{\text{plan}}}(a_i, b_i) = P_{i,t}^{\text{actual}}$, computed by binary search over the Beta CDF. Unlike classical SPI = BCWP/BCWS, which freezes once BCWS reaches BAC_i at the planned finish, $\text{SPI}_{i,t} = \text{ES}_{i,t}/(t - s_i + 1)$ degrades continuously as real time passes without progress, correctly distinguishing a project one period late from one three periods late (Lipke, 2003). This correction is adopted throughout the environment; classical SPI is not used.

$\text{SPI}_{i,t} < 1$ signals schedule lag; $\text{CPI}_{i,t} < 1$ signals cost overrun. $\text{EAC}_{i,t}$ is the forward-looking Estimate at Completion: it projects the total cost of the project at the current cost performance rate by scaling remaining planned work by the inverse of $\text{CPI}_{i,t}$, directly analogous to the forecast finish date computation used in Condition 1 of the termination mechanic (Section 3.2.5). Christensen (1993) established that this formulation produces the most accurate and stable cost-at-completion forecasts, with reliability increasing beyond the 20% completion mark (Lipke, 2003). The To-Complete Performance Index $\text{TCPI}_{i,t}$ (**??**) complements $\text{CPI}_{i,t}$ with a forward-looking cost signal: it measures the cost efficiency that must be sustained on remaining work to finish within BAC_i. $\text{TCPI}_{i,t} > 1$ indicates that future performance must exceed historical performance; values above 1.10 are widely considered unrecoverable in practice (Fleming and Koppelman, 2010; Christensen, 1993). While $\text{CPI}_{i,t}$ reflects what has happened, $\text{TCPI}_{i,t}$ reflects what must happen — the two together give the agent a complete cost picture at each period. All four indices are observable state variables; their stochastic dynamics at Levels 2–3 are calibrated from empirical EVM datasets (Acebes et al., 2014; Batselier and Vanhoucke, 2017).

3.2.5 Termination Mechanic

Termination is an endogenous environment mechanic, not an agent action. At each period t, the environment evaluates two conditions for every active project i. Both are forward-looking trajectory estimates, symmetric in structure to one another.

Condition 1 — Schedule tolerance exhausted. The forecast finish date, computed by linear extrapolation of remaining planned duration at the current schedule performance rate, is:

$$\hat{f}_{i,t} = t + \frac{D_i^{\text{plan}} - (t - s_i)}{\text{SPI}_{i,t}}. \tag{8}$$

The schedule deviation is $\Delta f_{i,t} = \hat{f}_{i,t} - f_i$. Condition 1 activates when the projected delay exceeds the contractual schedule tolerance Ω_i (expressed in periods):

$$\Delta f_{i,t} \; > \; \Omega_i. \tag{9}$$

Condition 2 — Cost overrun trajectory unacceptable. The cost deviation trajectory is defined symmetrically to $\Delta f_{i,t}$:

$$\Delta AC_{i,t} = \mathrm{EAC}_{i,t} - \mathrm{BAC}_i, \tag{10}$$

measuring the projected cost overrun in absolute cost units at the current performance rate. The Estimate at Completion $\mathrm{EAC}_{i,t}$ defined in (7) projects the total project cost at the current cost performance rate, by scaling remaining planned work by the inverse of $\mathrm{CPI}_{i,t}$ — directly analogous to the schedule extrapolation in (8). Condition 2 activates when the projected total cost exceeds the project-specific maximum tolerable cost overrun:

$$\Delta AC_{i,t} \;>\; \mu_i \cdot \mathrm{BAC}_i, \tag{11}$$

where $\mu_i \geq 0$ is the contractually set maximum cost overrun ratio for project i, fixed at commitment.

Cure-period counter and termination execution. When both conditions hold simultaneously, the environment decrements the project-specific cure-period counter, modelling the FIDIC Sub-Clause 15.1 obligation to allow a recovery interval before termination may be executed (Fédération Internationale des Ingénieurs-Conseils, 2017; Ndekugri and Rycroft, 2008):

$$\tau^{\mathrm{rem}}_{i,t} = \begin{cases} \tau^{\mathrm{rem}}_{i,t-1} - 1 & \text{if (9) and (11) both hold,} \\ \tau^{\mathrm{tol}}_i & \text{if either condition lapses.} \end{cases} \tag{12}$$

The counter initialises at $\tau^{\mathrm{rem}}_{i,s_i} = \tau^{\mathrm{tol}}_i$, where $\tau^{\mathrm{tol}}_i \in \mathbb{Z}^+$ is the project-specific tolerance, fixed at commitment. Termination executes at the period immediately following $t^{\mathrm{term}}_i = \min\{t : \tau^{\mathrm{rem}}_{i,t} = 0\}$. The counter is part of the observable portfolio state; the agent cannot override its dynamics but influences both conditions indirectly through allocation decisions that drive $\mathrm{SPI}_{i,t}$ and $\mathrm{CPI}_{i,t}$.

Upon termination at t^{term}_i, the net settlement payment is:

$$R^{\mathrm{term}}_i = P^{\mathrm{actual}}_{i,t^{\mathrm{term}}_i} \cdot CP_i \;-\; \left(A_i + \sum_{j \in \mathcal{M}^{\mathrm{cert}}_i(t^{\mathrm{term}}_i)} R^{\mathrm{net}}_{i,j} \right). \tag{13}$$

The first term is the contractor's entitlement for work physically delivered: the fraction of contract price proportional to actual progress at termination. The second term is the cumulative inflows already received: the advance payment A_i plus all net milestone payments $R^{\mathrm{net}}_{i,j}$ certified up to and including t^{term}_i, where $R^{\mathrm{net}}_{i,j}$ is as defined in (3). The settlement is the difference between what the contractor is owed for delivered scope and what they have already collected.

Positive R^{term}_i arises when delivered work exceeds cumulative receipts — typically when termination occurs just below a milestone threshold, so the contractor has performed work not yet captured in a certified payment. Negative R^{term}_i constitutes a net contractor obligation — typically when a large advance has been disbursed against work that was not subsequently delivered, or when certified milestone payments have run ahead of physical progress. In either case the settlement is fully determined by contract parameters and the progress level at termination; no penalty term or external calibration is required. This grounds the model's financial signals in the real asset value of work in place (Dixit and Pindyck, 1994) rather than in arbitrarily weighted deviation terms.

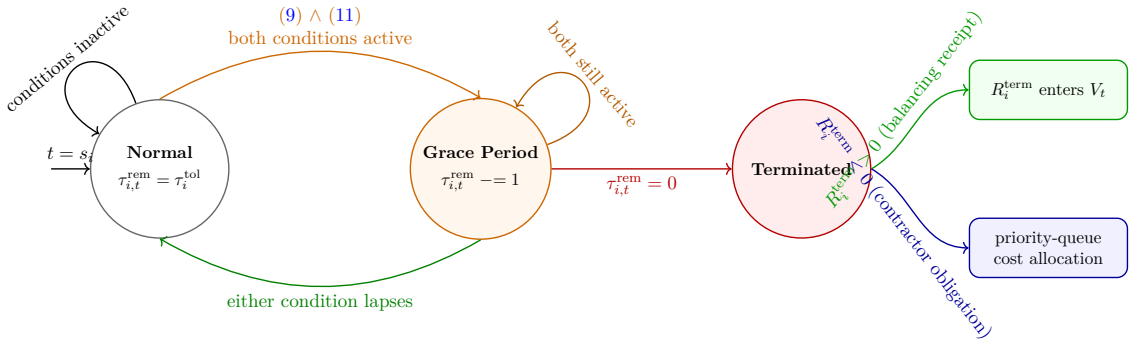


Figure 4: Finite-state representation of the termination mechanic for project i.

3.2.6 Portfolio Cash Balance

The portfolio cash balance evolves as:

$$B_{t+1} = B_t - \sum_{i \in \mathcal{P}_t^{\text{active}}} x_{i,t} + V_t. \tag{14}$$

Under the relaxed budget regime ($\kappa \gg 1$), the liquidity constraint $B_t \geq 0$ is non-binding by construction throughout all experiments. The shared balance B_t is the sole coupling across otherwise independent project dynamics and is retained in the formulation for completeness; it becomes binding in the tight and scarce regimes ($\kappa \approx 1$ and $\kappa \ll 1$) studied in sensitivity analysis.

3.2.7 Observable State and Shared Objective

The portfolio state at period t is:

$$S_t = \Big(\big\{ P^{\text{actual}}_{i,t},\ \text{ACWP}_{i,t},\ \text{SPI}_{i,t},\ \text{CPI}_{i,t},\ \text{TCPI}_{i,t},\ \text{EAC}_{i,t},\ \Delta f_{i,t},\ \text{status}_{i,t},\ \tau^{\text{rem}}_{i,t} \big\}_{i \in \mathcal{P}},\ B_t \Big), \tag{15}$$

where $\text{status}_{i,t} \in \{\texttt{active}, \texttt{completed}, \texttt{terminated}\}$. This gives a state dimension of $9n + 1$ per period.. All components are observable. One primitive is unobservable at decision time: $\eta_{i,t}$, realised only after allocation $x_{i,t}$ is committed. It is not part of the state vector; it is an environment-internal draw that the policy must act without knowing in advance.

The decision-maker maximises discounted net cash flow over the horizon:

$$\max_{\{x_{i,t}\}} \quad \mathbb{E}\left[\sum_{t=1}^{H} \gamma^{t-1} \left(V_t - \sum_{i \in \mathcal{P}_t^{\text{active}}} x_{i,t} \right) \right], \tag{16}$$

subject to, for all $i \in \mathcal{P}$, $t \in \mathcal{T}$:

$$x_{i,t} \geq 0, \tag{17}$$

$$\sum_{i \in \mathcal{P}_t^{\text{active}}} x_{i,t} \leq B_t, \tag{18}$$

$$x_{i,t} = 0 \quad \text{if } \text{status}_{i,t} \neq \texttt{active}. \tag{19}$$

Constraint (18) is retained for generality; under the relaxed budget regime it is non-binding. It becomes the operative constraint in the tight and scarce budget regimes.

There is no terminal sum and no deviation penalty in the objective. Completion incentives are embedded structurally: R_i^{ret} and R_{i,M_i}^{net} enter V_t at the final certification period; R_i^{term} (possibly negative) enters at termination, creating a natural deterrent against allowing project performance to deteriorate to the termination threshold (Ng et al., 1999). Symmetrically, the discounting of inflows by γ^{t-1} in (16) creates a completion incentive from above — earlier certification of milestone j yields $R_{i,j}^{\text{net}}$ at a higher discount weight, so a return-maximising agent will front-load allocations to projects where earlier certification is feasible, without any additional objective term. In competitive project-based markets where π_i is fixed at contract signing, this discounting effect is the formal expression of the capital velocity argument: the annualised return on committed capital increases with completion speed, and γ encodes the opportunity cost of capital that makes this concrete (Damodaran, 2012). The expectation $\mathbb{E}[\cdot]$ collapses to a deterministic evaluation at Level 1 and is retained at Levels 2 and 3.

3.3 Consolidated Parameter Reference

Table 2 lists all project parameters fixed at commitment, grouped by function. The planned S-curve and advance recovery parameters introduced in Sections 3.2.2–3.2.3 are included for completeness. The discount factor γ and initial balance B_1 are portfolio-level parameters shared across all projects.

3.4 Level 1: Deterministic Full-Foresight MILP

Setting $\eta_{i,t} = \bar{\eta}_{i,t}$ (known) renders S_t a deterministic function of $\mathbf{x}_t$, so $\mathcal{P}_t^{\text{active}}$, milestone certification times, and termination periods t_i^{term} are all computable ex ante. Under deterministic efficiency, the trajectories of $\text{SPI}_{i,t}$, $\text{CPI}_{i,t}$, $\text{EAC}_{i,t}$, and the counter $\tau_{i,t}^{\text{rem}}$ are fully determined before solving; both termination conditions (9)–(11) and the counter dynamics (12) are parameter-verifiable, so termination periods t_i^{term} and their settlements R_i^{term} enter the MILP as fixed parameters rather than optimisation constraints.

Introducing the certification binary $u_{i,j,t} \in \{0,1\}$ (= 1 iff milestone j of project i certified by period t) and activity indicator $\text{act}_{i,t} \in \{0,1\}$ (fixed parameter), the stochastic objective (16) collapses to:

Parameter	Symbol	Domain	Definition
Schedule			
Planned start & finish	$s_i,\ f_i$	$\mathbb{Z}^+$	Anchor of execution window and S-curve.
Planned duration	D_i^{plan}	$\mathbb{Z}^+$	$f_i - s_i + 1$; normalises progress against plan.
S-curve shape	$a_i,\ b_i$	$\mathbb{R}^+$	Beta CDF parameters governing planned expenditure profile (2).
Milestone threshold	$\theta_{i,j}$	$(0, 1]$	Cumulative progress fraction required to certify milestone j.
Earliest certification	$e_{i,j}$	$\mathbb{Z}^+$	Earliest period at which milestone j may be certified, regardless of progress achieved; enforces FIDIC Sub-Clause 14.3 payment interval constraints (Fédération Internationale des Ingénieurs-Conseils, 2017).
Schedule tolerance	Ω_i	$\mathbb{Z}^+$	Maximum admissible forecast finish overrun in periods before Condition 1 (9) activates.
To-Complete Performance Index	$\text{TCPI}_{i,t}$	$\mathbb{R}^+$	Forward-looking cost efficiency required on remaining work to finish within BAC_i; computed as $(\text{BAC}_i - \text{EV}_{i,t})/(\text{BAC}_i - \text{ACWP}_{i,t})$. Complements $\text{CPI}_{i,t}$ (historical) with a prospective signal; values above 1.10 are widely considered unrecoverable (Fleming and Koppelman, 2010).
Contract			
Contract price	CP_i	$\mathbb{R}^+$	Total client payment; $CP_i = \text{BAC}_i(1+\pi_i)$; payment accumulation baseline only.
Budget at completion	BAC_i	$\mathbb{R}^+$	Total planned contractor expenditure; EVM cost basis and progress normalisation denominator.
Profit margin	π_i	$\mathbb{R}_0^+$	Contractual markup on BAC_i.
Advance payment ratio	α_i	$[0, 1]$	$A_i = \alpha_i \cdot CP_i$ received at $t = s_i$.
Recovery deduction rate	δ_i^{rec}	$(0, 1]$	Fraction of each interim payment deducted toward advance recovery once threshold ψ_i is exceeded; default 0.25 per FIDIC Sub-Clause 14.2 (Fédération Internationale des Ingénieurs-Conseils, 2017).
Recovery threshold	ψ_i	$[0, 1]$	Cumulative certified payments as fraction of CP_i at which advance recovery begins; default 0.10 per FIDIC Sub-Clause 14.2 (Fédération Internationale des Ingénieurs-Conseils, 2017).
Retention ratio	ρ_i	$[0, 1]$	Fraction withheld per milestone; released as $\rho_i \cdot CP_i$ at completion.
Milestone weight	$\phi_{i,j}$	$\mathbb{R}^+$	Fraction of CP_i at milestone j; $\sum_j \phi_{i,j} = 1$.
Termination			
Max cost overrun ratio	μ_i	$\mathbb{R}_0^+$	Contractual ceiling on projected total cost expressed as fractional overrun on BAC_i; activates Condition 2 (11).
Termination tolerance	τ_i^{tol}	$\mathbb{Z}^+$	Cure-period length in periods; models FIDIC Sub-Clause 15.1 (Fédération Internationale des Ingénieurs-Conseils, 2017; Ndekugri and Rycroft, 2008).
Portfolio			
Discount factor	γ	$(0, 1]$	Per-period discount factor; encodes time

$$\max \quad \sum_{t=1}^{H} \gamma^{t-1} \left(V_t^{\text{det}} - \sum_i \text{act}_{i,t}\, x_{i,t} \right) \tag{20a}$$

$$\text{s.t.} \quad B_{t+1} = B_t - \sum_i \text{act}_{i,t}\, x_{i,t} + V_t^{\text{det}}, \qquad \forall\, t \tag{20b}$$

$$\sum_i \text{act}_{i,t}\, x_{i,t} \le B_t, \qquad \forall\, t \tag{20c}$$

$$P_{i,t}^{\text{actual}} = P_{i,t-1}^{\text{actual}} + \bar{\eta}_{i,t}\, \tfrac{x_{i,t}}{\text{BAC}_i}, \quad P_{i,t}^{\text{actual}} \le 1, \qquad \forall\, i,t \tag{20d}$$

$$P_{i,t}^{\text{actual}} \ge \theta_{i,j}\, u_{i,j,t} - \mathbf{M}(1 - u_{i,j,t}), \qquad \forall\, i,j,t \tag{20e}$$

$$P_{i,t}^{\text{actual}} \le \theta_{i,j} + \mathbf{M}\, u_{i,j,t}, \quad u_{i,j,t} \ge u_{i,j,t-1}, \quad u_{i,j+1,t} \le u_{i,j,t}, \qquad \forall\, i,j,t \tag{20f}$$

$$u_{i,j,t} = 0 \quad \text{if } t < e_{i,j}, \qquad \forall\, i,j,t \tag{20g}$$

$$V_t^{\text{det}} = \sum_{i,j} R_{i,j}^{\text{net}} \left(u_{i,j,t} - u_{i,j,t-1} \right) + \sum_i R_i^{\text{ret}} \left(u_{i,M_i,t} - u_{i,M_i,t-1} \right) + \sum_i A_i \cdot \mathbf{1}\{t = s_i\} + \sum_{i \in \mathcal{P}^{\text{term}}} R_i^{\text{term}} \cdot \mathbf{1}\{t = t_i^{\text{term}}\}, \qquad \forall\, t \tag{20h}$$

$$x_{i,t} = 0 \quad \text{if } \text{act}_{i,t} = 0, \qquad \forall\, i,t \tag{20i}$$

$$x_{i,t} \ge 0, \quad u_{i,j,t} \in \{0,1\}, \quad u_{i,j,0} = P_{i,0}^{\text{actual}} = 0. \tag{20j}$$

The big-M constant is $\mathbf{M} = \max_{i,j} \theta_{i,j}$. Constraints (20e)–(20f) implement exact if-and-only-if certification logic with monotone, ordered binary activation. Constraint (20g) enforces the contractual earliest certification date: $u_{i,j,t}$ is fixed to zero for all periods prior to $e_{i,j}$, preventing the MILP from certifying a milestone before its contractual window opens regardless of how much budget has been allocated. This makes the optimal allocation timing uniquely determined under $\gamma < 1$: the agent will front-load spend to reach $\theta_{i,j}$ exactly at $e_{i,j}$, then hold until the window opens. The difference $(u_{i,j,t} - u_{i,j,t-1})$ in (20h) equals one in the sole period a milestone is first certified, linearising the payment trigger without additional binaries.

Upper bound. Let Z_{L1}^* denote the Level 1 optimum. For any causal policy π feasible under uncertainty, $\sup_\pi \mathbb{E}[Z^\pi] \le Z_{\text{L1}}^*$: every feasible causal allocation sequence is also feasible in the Level 1 program solved on the realised trajectory, so the full-foresight optimum dominates in expectation. The gap $Z_{\text{L1}}^* - Z^\pi$ quantifies the cost of uncertainty and anchors the empirical comparison in Section **??**.

3.5 Level 2: Stochastic Extension and Intractability

3.5.1 Stochastic Primitive and Distributional Assumption

At Level 2, the deterministic efficiency scalar $\bar{\eta}_{i,t}$ is replaced by a random variable $\eta_{i,t}$ defined on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. The efficiency realisations are assumed to satisfy:

Assumption 1 (Efficiency structure)**.** For each project $i \in \mathcal{P}$ and period $t \in \mathcal{T}$:

(i) $\eta_{i,t} > 0$ almost surely;

(ii) $\{\eta_{i,t}\}_{t \in \mathcal{T}}$ are independent across periods given project i;

(iii) $\eta_{i,t}$ are independent across projects $i \in \mathcal{P}$;

(iv) the marginal distribution of $\eta_{i,t}$ is project-specific and calibrated from empirical EVM datasets (Acebes et al., 2014; Batselier and Vanhoucke, 2017).

The realisation $\eta_{i,t}(\omega)$ is revealed only after allocation $x_{i,t}$ is committed. This single timing constraint defines the one decision point at which the portfolio manager faces irreducible uncertainty: the allocation decision under unknown productivity. Under Relaxation B, payments are received in the period of certification, so no second stochastic primitive is required.

3.5.2 Information Structure and Non-Anticipativity

Let $\mathcal{F}_t$ denote the σ-algebra generated by all efficiency realisations and the resulting state transitions, up to and including period t:

$$\mathcal{F}_t = \sigma\big(\eta_{i,\tau}(\omega) : i \in \mathcal{P},\ \tau \le t\big), \qquad \mathcal{F}_0 = \{\emptyset, \Omega\}. \tag{21}$$

The history available to the portfolio manager at the start of period t, prior to committing $\mathbf{x}_t$, is:

$$\mathcal{H}_t = \big(S_1,\, \mathbf{x}_1,\, S_2,\, \mathbf{x}_2,\, \ldots,\, S_{t-1},\, \mathbf{x}_{t-1},\, S_t\big), \tag{22}$$

where S_t is the portfolio state defined in (15), which is $\mathcal{F}_{t-1}$-measurable since it reflects only realisations up to period $t-1$. The *non-anticipativity constraint* requires that the allocation at period t depends only on the available history:

$$\mathbf{x}_t \in \sigma(\mathcal{H}_t) \qquad \forall\, t \in \mathcal{T}. \tag{23}$$

This constraint distinguishes a causal policy from the full-foresight MILP of Level 1, which optimises over the entire realised trajectory $(\eta_{i,t})_{i,t}$ simultaneously.

3.5.3 The Multi-Stage Stochastic Program

Under Assumption 1 and constraint (23), the Level 1 MILP (20) becomes a multi-stage stochastic program (Birge and Louveaux, 2011). The decision at each stage is a *policy* $\pi_t : \mathcal{H}_t \to \mathbb{R}_{\ge 0}^{|\mathcal{P}_t^{\text{active}}|}$ rather than a fixed scalar, and the objective is an expectation over all scenario realisations. The program is:

$$\max_{\{\pi_t\}} \quad \mathbb{E}\left[\sum_{t=1}^{H} \gamma^{t-1}\left(V_t - \sum_{i \in \mathcal{P}_t^{\text{active}}} x_{i,t}\right)\right] \tag{24a}$$

$$\text{s.t.} \quad x_{i,t} = \pi_t(\mathcal{H}_t)_i, \qquad \mathbf{x}_t \in \sigma(\mathcal{H}_t), \qquad \forall\, i,\, t \tag{24b}$$

$$P_{i,t}^{\text{actual}} = P_{i,t-1}^{\text{actual}} + \eta_{i,t} \cdot \frac{x_{i,t}}{\text{BAC}_i}, \qquad \forall\, i,\, t \tag{24c}$$

$$B_{t+1} = B_t - \sum_{i \in \mathcal{P}_t^{\text{active}}} x_{i,t} + V_t, \qquad \forall\, t \tag{24d}$$

$$\sum_{i \in \mathcal{P}_t^{\text{active}}} x_{i,t} \le B_t, \qquad \forall\, t \tag{24e}$$

$$x_{i,t} = 0 \quad \text{if status}_{i,t} \neq \texttt{active}, \qquad \forall\, i,\, t \tag{24f}$$

$$x_{i,t} \ge 0, \qquad \forall\, i,\, t \tag{24g}$$

where V_t is as defined in (4), now a random variable whose realisation depends on $(\eta_{i,\tau})_{\tau \le t}$ through the progress recursion (24c) and the milestone certification events it generates.

The termination counter (12), and therefore $\mathcal{P}_t^{\text{active}}$ and $\text{status}_{i,t}$, are likewise random variables endogenous to the realised efficiency trajectory, in contrast to Level 1 where they were fixed parameters.

3.5.4 Scenario-Tree Representation and Exponential Growth

The canonical solution method for multi-stage stochastic programs is scenario-tree enumeration (Birge and Louveaux, 2011). The scenario tree for (24) has the following structure. Discretise $\eta_{i,t}$ to K values per project per period. At each period t, the tree branches by K^n (one branching factor of K per project, with n projects independent by Assumption 1(iii)). Over a horizon of H periods the tree has:

$$|\mathcal{S}| = K^{nH} \tag{25}$$

leaf nodes, each corresponding to a complete realisation of $(\eta_{i,t})_{i\in\mathcal{P},\, t\in\mathcal{T}}$.

For a representative infrastructure portfolio with $n = 10$ projects, a horizon H derived endogenously from the committed portfolio (typically $H \in [18, 36]$ months in practice), and $K = 5$ discrete efficiency scenarios per project per period, the leaf-node count is:

$$|\mathcal{S}| = 5^{10\times 24} = 5^{240} \approx 10^{167}, \tag{26}$$

a number that exceeds the estimated number of atoms in the observable universe by more than 80 orders of magnitude. This is not a consequence of the binary milestone variables inherited from the MILP structure: replacing them with continuous relaxations leaves the exponent nH unchanged, since the exponential growth is intrinsic to the multi-stage stochastic formulation and independent of the within-stage problem structure.

Endogenous horizon compounding. A further compounding factor distinguishes this problem from standard multi-stage stochastic programs with fixed horizons. The planning horizon $H = \max_i f_i - \min_i s_i + 1$ is itself a function of the committed portfolio; it is not an external configuration parameter. Under stochastic efficiency, the effective execution window of each project is uncertain — a project with $\text{SPI}_{i,t} < 1$ throughout will overrun f_i, extending the realised horizon beyond H. In the worst case, every project overruns simultaneously, and the actual number of periods over which decisions must be made grows with the portfolio. Each additional period multiplies the scenario tree by K^n, so horizon uncertainty compounds (25) multiplicatively.

Termination as structural bifurcation. At Level 1, termination periods t_i^{term} were fixed parameters precomputed from the known efficiency trajectory. At Level 2, the period at which the cure-period counter (12) reaches zero depends on the realised sequence of $\text{SPI}_{i,t}$ and $\text{CPI}_{i,t}$ values, which are in turn determined by $(\eta_{i,\tau})_{\tau\le t}$. Each termination event removes project i from $\mathcal{P}_t^{\text{active}}$ and delivers R_i^{term} into V_t — a structural change in the active project set that bifurcates the scenario tree at every period in which a project's counter is positive. Across n projects each with counter depth τ_i^{tol}, the number of structurally distinct subtrees grows combinatorially in the termination event space, adding a second source of exponential scaling beyond (25).

3.5.5 Intractability and the Motivation for Level 3

The exponential scaling established in (25) constitutes a fundamental computational barrier. It is not a function of solver technology or implementation efficiency; it is intrinsic

to the multi-stage stochastic formulation of any sequential decision problem with independent per-stage uncertainty across n components over H stages (Birge and Louveaux, 2011). Approximation methods for multi-stage stochastic programs — Sample Average Approximation (Kleywegt et al., 2002), Progressive Hedging (Rockafellar and Wets, 1991), and Stochastic Dual Dynamic Programming (Pereira and Pinto, 1991) — reduce the computational burden but do not escape the exponential scaling in n and H at portfolio scales of practical interest. Sample Average Approximation, for instance, requires a sample size that grows with the variance of the objective and the number of stages to achieve a given optimality gap (Kleywegt et al., 2002); for $n = 10$ and $H = 24$ this remains intractable without structural decomposition that the shared cash balance (24d) prevents.

Proposition 1 (Level 1 as upper bound)**.** *Let Z^*_{L1} denote the optimal value of the Level 1 MILP (20) evaluated on a realised efficiency trajectory $(\bar{\eta}_{i,t})_{i,t}$. Let Z^π denote the objective value achieved by any causal policy π satisfying (23) on the same trajectory. Then:*

$$\sup_{\pi \in \Pi^{\text{causal}}} \mathbb{E}[Z^\pi] \leq Z^*_{\text{L1}}, \tag{27}$$

where Π^{causal} denotes the set of all policies satisfying (23).

Proof. Fix any realised trajectory $\omega \in \Omega$. Any allocation sequence $\{x_{i,t}\}$ produced by a causal policy π under ω is a feasible solution to the Level 1 MILP solved on ω (since feasibility constraints (20b)– (20i) are identical to (24d)–(24f) for fixed $\bar{\eta}_{i,t} = \eta_{i,t}(\omega)$; the single uncertainty source means no separate payment delay trajectory need be fixed). Therefore $Z^\pi(\omega) \leq Z^*_{\text{L1}}(\omega)$ for every ω. Taking expectations on both sides yields (27). □

The gap $Z^*_{\text{L1}} - \mathbb{E}[Z^\pi]$ quantifies the cost of uncertainty: the value destroyed by operating under incomplete information relative to a hypothetical full-foresight oracle. This gap is the primary empirical quantity reported in Section **??**. The intractability of computing the exact solution to (24) motivates the Level 3 reformulation: rather than enumerating the scenario tree, the problem is recast as a sequential interaction between a policy and an environment, solved through sampled trajectory experience rather than exhaustive enumeration.

3.6 Level 3: Constrained MDP and RL Agent

The intractability established in Section 3.5 rules out exact solution of (24) at realistic portfolio scales. The Level 3 reformulation replaces exhaustive scenario-tree enumeration with a sequential interaction framework: the portfolio manager is modelled as a *policy* that observes the current state, commits an allocation, and receives a reward signal, with the environment transitioning stochastically to the next state. Optimisation proceeds through repeated sampled interaction rather than explicit enumeration, making the computational cost per training step independent of $|\mathcal{S}| = K^{nH}$.

3.6.1 Constrained Markov Decision Process Formulation

Definition 1 (Portfolio Governance CMDP)**.** The portfolio budget allocation problem at Level 3 is formalised as a finite-horizon Constrained Markov Decision Process (Altman, 1999):

$$\mathcal{M} = \big(\mathcal{S},\ \mathcal{A},\ \mathcal{T},\ R,\ \mathcal{C},\ \gamma,\ H\big), \tag{28}$$

with components defined as follows.

State space $\mathcal{S}$. The state at period t is the portfolio state S_t defined in (15):

$$S_t \in \mathcal{S} \subseteq [0,1]^n \times \mathbb{R}^n_{\geq 0} \times \mathbb{R}^{4n} \times \{0,1,2\}^n \times \mathbb{Z}^n_{\geq 0} \times \mathbb{R}_{\geq 0}, \tag{29}$$

collecting, for each project i: actual progress $P^{\text{actual}}_{i,t} \in [0,1]$, cumulative actual cost $\text{ACWP}_{i,t} \in \mathbb{R}_{\geq 0}$, performance indices $\text{SPI}_{i,t}$, $\text{CPI}_{i,t}$, $\text{TCPI}_{i,t}$, $\text{EAC}_{i,t}$, schedule deviation $\Delta f_{i,t}$, project status $\text{status}_{i,t} \in \{0,1,2\}$ encoding $\{\texttt{active}, \texttt{completed}, \texttt{terminated}\}$, and cure-period counter $\tau^{\text{rem}}_{i,t} \in \mathbb{Z}_{\geq 0}$; together with the portfolio cash balance $B_t \in \mathbb{R}_{\geq 0}$. The state dimension is $9n + 1$ per period. All components are observable. One primitive is unobservable at decision time: $\eta_{i,t}$, realised only after allocation $x_{i,t}$ is committed. It is not part of the state vector; it is an environment-internal draw that the policy must act without knowing in advance.

Action space $\mathcal{A}$ and feasible set. The raw action space is:

$$\mathcal{A} = \mathbb{R}^n_{\geq 0}, \tag{30}$$

with the state-dependent feasible action set:

$$\mathcal{A}(S_t) = \left\{ \mathbf{x}_t \in \mathbb{R}^n_{\geq 0} \;\middle|\; \sum_{i \in \mathcal{P}^{\text{active}}_t} x_{i,t} \leq B_t, \quad x_{i,t} = 0 \text{ if } \text{status}_{i,t} \neq \texttt{active} \right\}, \tag{31}$$

encoding constraints (18) and (19) directly. The hard liquidity constraint $B_t \geq 0$ is enforced by construction through the projection operator defined below.

Projection operator. To guarantee $\mathbf{x}_t \in \mathcal{A}(S_t)$ for any raw network output $\tilde{\mathbf{x}}_t \in \mathbb{R}^n$, the environment applies a two-step projection before executing the transition. First, allocations to inactive projects are zeroed: $\tilde{x}_{i,t} \leftarrow 0$ for all i with $\text{status}_{i,t} \neq \texttt{active}$. Second, the remaining vector is projected onto the budget simplex:

$$\mathbf{x}_t = B_t \cdot \frac{\max(\tilde{\mathbf{x}}^{\text{active}}_t, \mathbf{0})}{\left\| \max(\tilde{\mathbf{x}}^{\text{active}}_t, \mathbf{0}) \right\|_1 \vee 1}, \tag{32}$$

where $\tilde{\mathbf{x}}^{\text{active}}_t$ denotes the sub-vector restricted to active projects, and $\vee$ denotes the maximum operator ensuring the denominator is at least one when the agent outputs all-zero or all-negative values. This projection is differentiable almost everywhere and preserves gradient flow through the policy network during training (Schulman et al., 2017).

Transition kernel $\mathcal{T}$. The stochastic transition $\mathcal{T}(S_{t+1} \mid S_t, \mathbf{x}_t)$ is defined by the composition of environment mechanics applied sequentially within each period:

(i) Draw $\eta_{i,t} \sim F_i$ independently for each active project i, where F_i is the project-specific efficiency distribution calibrated from Acebes et al. (2014); Batselier and Vanhoucke (2017).

(ii) Update progress and actual cost via (1).

(iii) Compute EVM signals via (5)–(7).

(iv) Evaluate termination conditions (9)–(11) and update counter via (12).

(v) Certify milestones for which $P_{i,t}^{\text{actual}} \geq \theta_{i,j}$ *and* $t \geq e_{i,j}$. Each newly certified milestone (i,j) contributes $R_{i,j}^{\text{net}}$ to V_t immediately in the certification period (Relaxation B). Compute V_t via (3)–(4).

(vi) Update portfolio cash balance via (14).

(vii) Execute termination for any project with $\tau_{i,t}^{\text{rem}} = 0$ or $t \geq f_i + \Omega_i$, compute R_i^{term} via (13).

The transition kernel is Markovian: S_{t+1} depends on $(S_t, \mathbf{x}_t, \boldsymbol{\eta}_t)$ only, not on $\mathcal{H}_{t-1}$. This is the key structural property that makes the CMDP formulation exact rather than approximate: the full portfolio state S_t defined in (15) is a sufficient statistic for the history $\mathcal{H}_t$ defined in (22).

Reward signal R. The per-period reward is the discounted net cash flow term from (16):

$$r_t(S_t, \mathbf{x}_t, S_{t+1}) = \gamma^{t-1} \left(V_t - \sum_{i \in \mathcal{P}_t^{\text{active}}} x_{i,t} \right), \tag{33}$$

where V_t is realised after the transition. The reward contains no deviation penalty term; completion incentives are embedded structurally through R_i^{ret} and R_{i,M_i}^{net} at project completion and through R_i^{term} (possibly negative) at termination, as established in Section 3.2.7.

Constraint set $\mathcal{C}$. The hard liquidity constraint $B_t \geq 0$ is the sole constraint promoted to the CMDP constraint set:

$$\mathcal{C} = \{ \mathbf{x}_t \in \mathcal{A} \mid B_{t+1} \geq 0 \}. \tag{34}$$

This constraint is enforced by the projection (32) rather than through a Lagrangian penalty, ensuring hard satisfaction at every step rather than satisfaction only in expectation (Altman, 1999).

CMDP objective. The policy optimisation problem is:

$$\pi^* = \text{argmax}_{\pi \in \Pi} \; J(\pi) \; = \; \text{argmax}_{\pi \in \Pi} \; \mathbb{E}_\pi \left[\sum_{t=1}^{H} r_t(S_t, \mathbf{x}_t, S_{t+1}) \right], \tag{35}$$

subject to $\mathbf{x}_t \in \mathcal{C}$ for all t, where Π denotes the set of all measurable policies satisfying (23) and $\mathbb{E}_\pi$ denotes expectation under the distribution over trajectories induced by π and $\mathcal{T}$. The objective $J(\pi)$ in (35) is identical to the intractable stochastic program (24a); the CMDP reformulation does not change the objective but changes the solution method from scenario-tree enumeration to policy-gradient optimisation.

Remark 1 (Relationship to Level 1 upper bound). By Proposition 1, $J(\pi^*) \leq Z_{\text{L1}}^*$ for any causal policy π^*. The CMDP does not close this gap; it provides the tractable path to finding the best achievable $J(\pi)$ within Π^{causal}. The empirical gap $Z_{\text{L1}}^* - J(\hat{\pi})$, where $\hat{\pi}$ is the trained PPO agent, quantifies the combined cost of uncertainty and approximation error in Section **??**.

3.6.2 Gymnasium-Compatible Environment

The CMDP $\mathcal{M}$ is implemented as a configurable Gymnasium-compatible environment (Towers et al., 2024). The environment exposes the standard `reset()` / `step()` interface, with `observation_space` $= \mathcal{S}$ and `action_space` $= \mathcal{A}$. All project and portfolio parameters (Table 2) are passed as a configuration dictionary at initialisation; the planning horizon H is derived automatically as $\max_i f_i - \min_i s_i + 1$ from the committed project set, making the environment fully parameterisable across contract types and portfolio sizes without code modification.

The observation vector returned at each step is a flattened concatenation of S_t as defined in (29), normalised to $[0, 1]$ per component to facilitate stable policy gradient estimation. The environment enforces the projection (32) on the raw action vector before applying the transition kernel, guaranteeing $\mathbf{x}_t \in \mathcal{A}(S_t)$ regardless of the policy output. Episodes terminate when $t = H$ or when all projects have reached `completed` or `terminated` status, whichever occurs first.

The stochastic efficiency $\eta_{i,t}$ is drawn each period from a project-specific Beta distribution calibrated from empirical EVM datasets (Acebes et al., 2014; Batselier and Vanhoucke, 2017); the Beta family accommodates the right-skewed, bounded support structure of empirical CPI and SPI distributions documented in Lipke (2003) and Batselier and Vanhoucke (2017). Under Relaxation B, milestone payments are received in the certification period; no payment delay distribution is sampled.

3.6.3 PPO Agent and Training Protocol

Algorithm selection. Proximal Policy Optimisation (Schulman et al., 2017) is selected as the solution algorithm for three reasons. First, its clipped surrogate objective provides stable updates in continuous action spaces without the step-size sensitivity of vanilla policy gradient methods. Second, its on-policy sample collection is compatible with the non-stationary episode structure of the environment, in which the active project set $\mathcal{P}_t^{\text{active}}$ changes endogenously across periods. Third, PPO has demonstrated consistent performance in operations management domains with similar state-action structure (Giannoccaro and Pontrandolfo, 2002; Peng and Huang, 2021).

PPO surrogate objective. At each training iteration, PPO collects a batch of trajectories $\mathcal{D} = \{\tau_k\}_{k=1}^{N}$ under the current policy π_θ and optimises the clipped surrogate:

$$\mathcal{L}^{\text{CLIP}}(\theta) = \mathbb{E}_{\mathcal{D}}\left[\min\left(\frac{\pi_\theta(a_t \mid s_t)}{\pi_{\theta_{\text{old}}}(a_t \mid s_t)}\hat{A}_t,\ \text{clip}\left(\frac{\pi_\theta(a_t \mid s_t)}{\pi_{\theta_{\text{old}}}(a_t \mid s_t)},\, 1-\epsilon,\, 1+\epsilon\right)\hat{A}_t\right)\right], \tag{36}$$

where $\hat{A}_t$ is the Generalised Advantage Estimate (Schulman et al., 2016) computed from the value function baseline $V_\phi(S_t)$ trained jointly with the policy, and ϵ is the clipping threshold. The value function shares the lower network layers with the policy (actor-critic architecture), with separate output heads for π_θ and V_ϕ.

Policy architecture. The policy network maps the observation vector $S_t \in \mathbb{R}^{8n+1}$ to a raw allocation vector $\tilde{\mathbf{x}}_t \in \mathbb{R}^n$ through a fully connected network with layer normalisation. The raw output is passed through the projection (32) before being applied to the environment, ensuring constraint satisfaction without a penalty term in (36). The output distribution is a diagonal Gaussian with state-dependent mean $\mu_\theta(S_t)$ and learned log-standard-deviation $\log \sigma_\theta$, consistent with standard continuous-action PPO implementations.

Domain randomisation. To ensure the learned policy $\hat{\pi}$ generalises across project types, contract structures, and market conditions rather than overfitting to a single parameterisation, all parameters in Table 2 are randomised within calibrated ranges at the start of each training episode (Tobin et al., 2017). Formally, at each episode reset, a parameter vector $\boldsymbol{\xi} \sim \mathcal{U}(\boldsymbol{\Xi})$ is sampled from a calibrated uniform distribution over the parameter space $\boldsymbol{\Xi}$, and the environment is initialised with $\boldsymbol{\xi}$. The trained policy $\hat{\pi}$ implicitly conditions on the observable consequences of $\boldsymbol{\xi}$ (as reflected in S_t) rather than on $\boldsymbol{\xi}$ directly, producing a policy that is robust to parameter uncertainty without requiring an explicit model of the parameter distribution at deployment (Tobin et al., 2017).

Validation protocol. Walk-forward validation (Cerqueira et al., 2020) is used as the empirical evaluation protocol to guard against data leakage given the time-series structure of project cash-flow data. The training set consists of episodes whose parameter draws fall within the calibrated historical range; the test set consists of held-out parameter configurations drawn from the tails of $\boldsymbol{\Xi}$, representing out-of-distribution contract and market conditions. All performance metrics in Section **??** are reported on the held-out test set.

3.6.4 Evaluation Baselines

The trained agent $\hat{\pi}$ is evaluated against three baselines that span the tractability-optimality spectrum established by the three-level arc.

Baseline 1 — Level 1 MILP oracle. The MILP (20) is solved on each test episode's realised efficiency trajectory $(\eta_{i,t}(\omega))_{i,t}$ using a commercial solver. Its objective value $Z^*_{\text{L1}}(\omega)$ is the per-episode upper bound from Proposition 1. The gap $Z^*_{\text{L1}} - J(\hat{\pi})$ is the primary performance metric.

Baseline 2 — Two-stage stochastic program. A two-stage approximation of (24) is solved by fixing first-stage decisions $\mathbf{x}_1$ and optimising over the expected recourse across N_s sampled scenarios for subsequent stages (Birge and Louveaux, 2011). This represents the best tractable exact method available within the stochastic programming tradition and quantifies the value of the full sequential policy over a rolling two-stage approximation.

Baseline 3 — Proportional heuristic. At each period t, the available balance B_t is distributed across active projects proportionally to their remaining planned expenditure $\text{BAC}_i(1 - P^{\text{actual}}_{i,t})$:

$$x^{\text{prop}}_{i,t} = B_t \cdot \frac{\text{BAC}_i(1 - P^{\text{actual}}_{i,t})}{\displaystyle\sum_{j \in \mathcal{P}^{\text{active}}_t} \text{BAC}_j(1 - P^{\text{actual}}_{j,t})}. \tag{37}$$

This rule is parameter-free and represents current practice in portfolio management information systems: budget is distributed in proportion to remaining work without regard to contract profitability, schedule urgency, or termination risk. The gap between the PPO agent and this baseline quantifies the practical value of the learned policy over the incumbent approach.

References

Acebes, F., Pajares, J., Galán, J. M., and López-Paredes, A. (2014). A new approach for project control under uncertainty: Going back to the basics. *International Journal of Project Management*, 32(3):423–434.

Altman, E. (1999). *Constrained Markov Decision Processes*. Routledge, New York, 1st edition.

Archer, N. P. and Ghasemzadeh, F. (1999). An integrated framework for project portfolio selection. *International Journal of Project Management*, 17(4):207–216.

Arditi, D. and Chotibhongs, R. (2005). Issues in subcontracting practice. *Journal of Construction Engineering and Management*, 131(8):866–876.

Asghari, V., Biglari, A. J., and Hsu, S.-C. (2023). Multiagent Reinforcement Learning for Project-Level Intervention Planning under Multiple Uncertainties. *Journal of Management in Engineering*, 39(2):04022075.

Bahroun, Z., Tanash, M., As'ad, R., and Alnajar, M. (2023). Artificial Intelligence Applications in Project Scheduling: A Systematic Review, Bibliometric Analysis, and Prospects for Future Research. *Management Systems in Production Engineering*, 31:144–161.

Bai, L., Song, C., Zhou, X., Tian, Y., and Wei, L. (2023). Assessing project portfolio risk via an enhanced ga-bpnn combined with pca. *Engineering Applications of Artificial Intelligence*, 126:106779.

Baqeri, K., Mohammadi, E., and Gilani, M. (2019). Multi objective project portfolio selection. *Journal of Project Management*, pages 249–256.

Barraza, G. A., Back, W. E., and Mata, F. (2004). Probabilistic monitoring of project performance using S-curves. *Journal of Construction Engineering and Management*, 130(1):142–148.

Batselier, J. and Vanhoucke, M. (2017). Improving project forecast accuracy by integrating earned value management with exponential smoothing and reference class forecasting. *International Journal of Project Management*, 35(1):28–43.

Beseiso, M. and Kumar, G. (2021). A fuzzy computational approach for selecting interdependent projects using prioritized criteria. *Journal of Intelligent & Fuzzy Systems*, 40(6):11341–11354.

Birge, J. R. and Louveaux, F. (2011). *Introduction to Stochastic Programming*. Springer, New York, 2nd edition.

Cerqueira, V., Torgo, L., and Mozetič, I. (2020). Evaluating time series forecasting models: An empirical study on performance estimation methods. *Machine Learning*, 109:1997–2028.

Christensen, D. S. (1993). The estimate at completion problem: A review of three studies. *Project Management Journal*, 24(1):37–42.

Christensen, D. S. and Heise, S. R. (1993). Cost performance index stability. *National Contract Management Journal*, 25:7–15.

Cooper, R. G., Edgett, S. J., and Kleinschmidt, E. J. (2001). *Portfolio Management for New Products*. Perseus Publishing, Cambridge, MA, 2nd edition.

Damodaran, A. (2012). *Investment Valuation: Tools and Techniques for Determining the Value of Any Asset*. Wiley Finance. John Wiley & Sons, Hoboken, NJ, 3rd edition.

Dixit, A. K. and Pindyck, R. S. (1994). *Investment under Uncertainty*. Princeton University Press, Princeton, NJ.

Doloi, H. (2012). Cost overruns and failure in project management: Understanding the roles of key stakeholders in construction projects. *Journal of Construction Engineering and Management*, 138(3):267–279.

Ebnerasoul, M., Ghannadpour, S. F., and Haeri, A. (2023). A collective efficacy-based approach for bi-objective sustainable project portfolio selection using interdependency network model between projects. *Environment, Development and Sustainability*, 25:13981–14001.

Eilat, H., Golany, B., and Shtub, A. (2006). Constructing and evaluating balanced portfolios of R&D projects with interactions: A DEA based methodology. *European Journal of Operational Research*, 172(3):1018–1039.

El-Sabek, L. M. and McCabe, B. Y. (2018). Coordination Challenges of Production Planning in the Construction of International Mega-Projects in The Middle East. *International Journal of Construction Education and Research*, 14(2):118–140.

Ennis, C. and Burr, A. (2021). Termination under FIDIC: Conditions, consequences and disputes. *International Construction Law Review*, 38(2):112–134.

Fédération Internationale des Ingénieurs-Conseils (2017). *Conditions of Contract for EPC/Turnkey Projects (Silver Book)*. FIDIC, Geneva, 2nd edition.

Fleming, Q. W. and Koppelman, J. M. (2010). *Earned Value Project Management*. Project Management Institute, Newtown Square, PA, 4th edition.

Flyvbjerg, B. (2017). *The Oxford Handbook of Megaproject Management*. Oxford University Press, Oxford.

Flyvbjerg, B., Holm, M. K. S., and Buhl, S. L. (2003). How common and how large are cost overruns in transport infrastructure projects? *Transport Reviews*, 23(1):71–88.

Ghapanchi, A. H., Tavana, M., Khakbaz, M. H., and Low, G. (2012). A methodology for selecting portfolios of projects with interactions and under uncertainty. *International Journal of Project Management*, 30(7):791–803.

Giannoccaro, I. and Pontrandolfo, P. (2002). Inventory management in supply chains: A reinforcement learning approach. *International Journal of Production Economics*, 78(2):153–161.

Huchzermeier, A. and Loch, C. H. (2001). Project management under risk: Using the real options approach to evaluate flexibility in R&D. *Management Science*, 47(1):85–101.

Jiang, C., Li, X., Lin, J.-R., Liu, M., and Ma, Z. (2023). Adaptive control of resource flow to optimize construction work and cash flow via online deep reinforcement learning. *Automation in Construction*, 150:104817.

Jiang, Y., Olmo, J., and Atwi, M. (2024). Deep reinforcement learning for portfolio selection. *Global Finance Journal*, 62:101016.

Kaka, A. P. and Price, A. D. F. (1993). Modelling standard cost commitment curves for contractors' cash flow forecasting. *Construction Management and Economics*, 11(4):271–283.

Karimi, S., Saeed, M., and Pishvaee, M. (2021). A scenario-based mathematical approach to a robust project portfolio selection problem under fuzzy uncertainty. *Journal of Intelligent & Fuzzy Systems*, 42:1–14.

Kavadias, S. and Loch, C. H. (2003). *Project Selection under Uncertainty: Dynamically Allocating Resources to Maximize Value*. Kluwer Academic Publishers, Boston, MA.

Kenley, R. and Wilson, O. (2009). *A Construction Management Approach to S-Curves and Cash Flow Management*. Spon Press, London.

Killen, C. P., Hunt, R. A., and Kleinschmidt, E. J. (2008). Project portfolio management for product innovation. *International Journal of Quality & Reliability Management*, 25(1):24–38.

Kim, E.-B. and Ballard, G. (2002). Case study: Using earned value method in field management of construction project. *Journal of Construction Engineering and Management*, 128(5):436–445.

Kleywegt, A. J., Shapiro, A., and Homem-de Mello, T. (2002). The sample average approximation method for stochastic discrete optimization. *SIAM Journal on Optimization*, 12(2):479–502.

Kosztyán, Z. T., Katona, A. I., Kuppens, K., Kisgyörgy-Pál, M., Nachbagauer, A., and Csizmadia, T. (2022). Exploring the structures and design effects of eu-funded r&d&i project portfolios. *Technological Forecasting and Social Change*, 180:121687.

Kwak, Y. H. and Ingall, L. (2007). Exploring Monte Carlo simulation applications for project management. *Risk Management*, 9(1):44–57.

Lipke, W. (2003). Schedule is different. *The Measurable News*, pages 10–15.

Liu, Z., Liang, Y., and Zheng, H. (2023). Multi-project scheduling based on deep reinforcement learning. *Automation in Construction*, 148:104721.

Lova, A., Maroto, C., and Tormos, P. (2000). A multicriteria heuristic method to improve resource allocation in multiproject scheduling. *European Journal of Operational Research*, 127(2):408–424.

Markowitz, H. (1952). Portfolio selection. *The Journal of Finance*, 7(1):77–91.

Martinsuo, M. and Lehtonen, P. (2007). Role of single-project management in achieving portfolio management efficiency. *International Journal of Project Management*, 25(1):56–65.

Mazhari, N. and Tajik, S. (2024). Evaluating the impact of artificial intelligence and machine learning in project portfolio management. Master's thesis, Stockholm University.

Merrow, E. W. (2011). *Industrial Megaprojects: Concepts, Strategies, and Practices for Success*. Wiley, Hoboken, NJ.

Mohagheghi, V., Mousavi, S. M., Mojtahedi, M., and Newton, S. (2020). Evaluating large, high-technology project portfolios using a novel interval-valued pythagorean fuzzy set framework: An automated crane project case study. *Expert Systems with Applications*, 162:113007.

Ndekugri, I. and Rycroft, M. (2008). The FIDIC engineering contracts: A commentary. *Construction Law Journal*, 24(3):181–198. Analysis of Sub-Clause 15.1 cure period obligations and wrongful termination jurisprudence.

Ng, A. Y., Harada, D., and Russell, S. (1999). Policy invariance under reward transformations: Theory and application to reward shaping. In *Proceedings of the 16th International Conference on Machine Learning (ICML)*, pages 278–287.

Noroozi, A., Mokhtari, H., Bagherpour, M., and Sadjadi, S. J. (2016). A probabilistic portfolio budget allocation problem with CPI index under risk. *Journal of Industrial and Production Engineering*, 33(4):236–246.

Page, M. J., McKenzie, J. E., Bossuyt, P. M., Boutron, I., Hoffmann, T. C., Mulrow, C. D., Shamseer, L., Tetzlaff, J. M., Akl, E. A., Brennan, S. E., Chou, R., Glanville, J., Grimshaw, J. M., Hróbjartsson, A., Lalu, M. M., Li, T., Loder, E. W., Mayo-Wilson, E., McDonald, S., McGuinness, L. A., Stewart, L. A., Thomas, J., Tricco, A. C., Welch, V. A., Whiting, P., and Moher, D. (2021). The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *BMJ*, 372:n71.

Pappert, L. and Kusanke, K. (2023). Modern Project Portfolio Management–Analyzing the Potential of Artificial Intelligence. In Axel Kalenborn et al., editor, *Projektmanagement und Vorgehensmodelle*, Lecture Notes in Informatics (LNI), page 125, Bonn. Gesellschaft für Informatik.

Park, M. and Peña-Mora, F. (2004). Reliability buffering for construction projects. *Journal of Construction Engineering and Management*, 130(5):626–637.

Peng, W. and Huang, M. (2021). A deep reinforcement learning approach for construction project scheduling. *Applied Sciences*, 11(3):1044.

Pereira, M. V. F. and Pinto, L. M. V. G. (1991). Multi-stage stochastic optimization applied to energy planning. *Mathematical Programming*, 52(1):359–375.

Pourahmadi, K., Nouri, S., and Yaghoubi, S. (2015). A scenario based project portfolio selection. *Management Science Letters*, pages 883–888.

Project Management Institute (2017). *The Standard for Portfolio Management*. Project Management Institute, Newtown Square, PA, 4th edition.

Project Management Institute (2021). *A Guide to the Project Management Body of Knowledge (PMBOK Guide)*. Project Management Institute, Newtown Square, PA, 7th edition.

Ranjbar, M., Nasiri, M. M., and Torabi, S. A. (2022). Multi-mode project portfolio selection and scheduling in a build-operate-transfer environment. *Expert Systems with Applications*, 189:116134.

Rockafellar, R. T. and Wets, R. J.-B. (1991). Scenarios and policy aggregation in optimization under uncertainty. *Mathematics of Operations Research*, 16(1):119–147.

Schulman, J., Moritz, P., Levine, S., Jordan, M., and Abbeel, P. (2016). High-dimensional continuous control using generalized advantage estimation. *arXiv preprint arXiv:1602.01783*.

Schulman, J., Wolski, F., Dhariwal, P., Radford, A., and Klimov, O. (2017). Proximal policy optimization algorithms. *arXiv preprint arXiv:1707.06347*.

Shou, Y. Y. and Huang, Y. L. (2010). Combinatorial auction algorithm for project portfolio selection and scheduling to maximize the net present value. *Journal of Zhejiang University. Science. C*, 11:562–574.

Staw, B. M. and Ross, J. (1987). Knowing when to pull the plug. *Harvard Business Review*, 65(2):68–74.

Sutton, R. S. and Barto, A. G. (2018). *Reinforcement Learning: An Introduction*. MIT Press, Cambridge, MA, 2nd edition.

Szwarcfiter, C., Herer, Y. T., and Shtub, A. (2023). Balancing Project Schedule, Cost, and Value under Uncertainty: A Reinforcement Learning Approach. *Algorithms*, 16(8):395.

Tassel, P., Gebser, M., and Schekotihin, K. (2023). An end-to-end reinforcement learning approach to job-shop scheduling problems. *Proceedings of the International Conference on Automated Planning and Scheduling*, 33:338–347.

Tobin, J., Fong, R., Ray, A., Schneider, J., Zaremba, W., and Abbeel, P. (2017). Domain randomization for transferring deep neural networks from simulation to the real world. In *Proceedings of the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 23–30.

Touran, A. (2006). A mathematical model for establishing budget levels for a portfolio of transit projects. *Proc., ICCCBE XI*.

Touran, A. (2010). Probabilistic Approach for Budgeting in Portfolio of Projects. *Journal of Construction Engineering and Management*, 136(3):361–370.

Towers, M., Terry, J. K., Kwiatkowski, A., et al. (2024). Gymnasium: A standard interface for reinforcement learning environments. *arXiv preprint arXiv:2407.17032*.

Turner, J. R. (2014). *The Handbook of Project-Based Management*. McGraw-Hill, New York, 4th edition.

Van Peteghem, V. and Vanhoucke, M. (2010). A genetic algorithm for the preemptive and non-preemptive multi-mode resource-constrained project scheduling problem. *European Journal of Operational Research*, 201(2):409–418.

Yamakawa, E. K., Cauchick-Miguel, P. A., Sousa-Zomer, T. T., and Killen, C. P. (2019). Project portfolio management: A landscape of the literature. *International Journal of Business Excellence*, 18(4):450–487.

Zhang, X., Qiu, Y., and Pardalos, P. M. (2020). Project portfolio selection and scheduling under uncertainty using multi-objective evolutionary algorithm. *Annals of Operations Research*, 285:3–24.

A Appendix A : PRISMA 2020 Flow Diagram

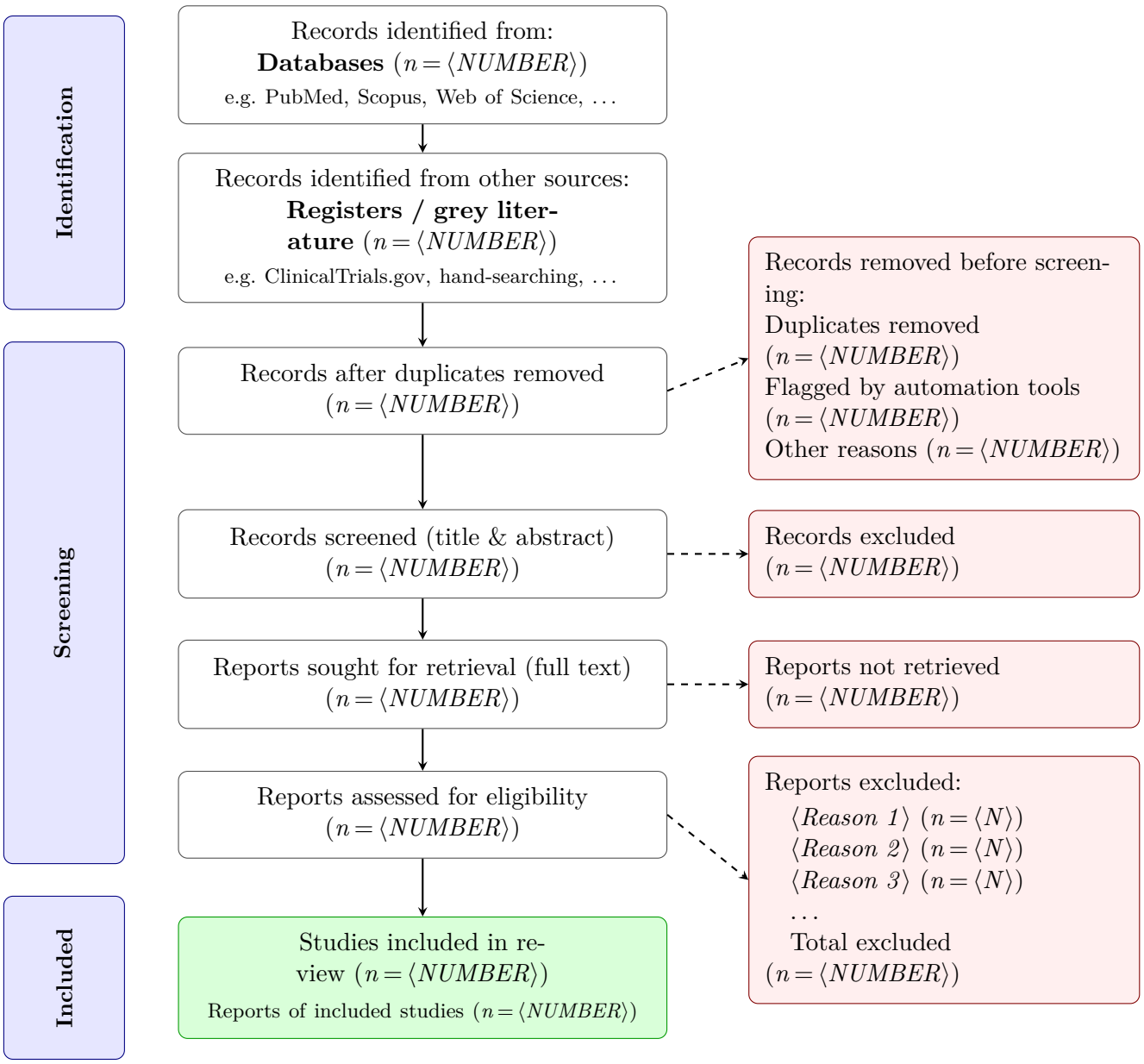


Figure 5: PRISMA 2020 flow diagram of the literature search and selection process.

A Appendix B : Environment verification suite

Master Figure Legends

All figures in this document use a consistent visual language. Legends are omitted from individual plots; refer to the key below.

Figure type	Symbol / line / colour meaning
EVM panel (left)	f_i (planned finish) BCWS (planned value) BCWP (earned value) ACWP (actual cost) $\theta_{i,j}$ (progress threshold)
	Red shaded areas (if present) indicate schedule/cost overrun caps.
Milestone profile (centre)	$x_{i,t}$ (budget allocation) $R^{\text{net}}_{i,j}$, A_i, R^{ret}_i (inflows)
	Yellow bands (if any) represent payment delay windows.
SPI / CPI panel (right)	SPI (•) CPI (■) reference = 1
Portfolio: cash balance (left)	B_t (cash balance) B_1 (initial cash)
Portfolio: period cash flows (centre)	Outflow Inflow Net (line with ○)
Portfolio: cumul. discounted NCF (right)	cumul. disc. NCF

All axes are scaled to the case horizon H (xmax $= H + 1$). The middle panel of every projectlevel figure uses a symmetric vertical range $[-\text{BAC}_i, \text{BAC}_i]$ to display allocations and inflows on a common scale. Portfolio figures use a range appropriate to the aggregated flows.

Conventions

All cases share these conventions unless explicitly overridden:

- Each discrete timestep t represents the interval from the end of period $t-1$ to the end of period t. All events assigned to period t (spend, progress update, milestone certification, payment) occur at the close of that interval.
- $\gamma = 0.95$ throughout, imposing time value of money in every case.
- Earliest certification dates $e_{i,j}$ are set for every case. The default rule is $e_{i,j} = s_i + \lfloor (j-1) \cdot D_i^{\text{plan}} / M_i \rfloor$, spreading milestones evenly across the planned duration. Any departure is stated explicitly.
- **Advance payment** is received at $t = s_i - 1$. For single-project cases where $s_i = 1$, this means the advance arrives at $t = 0$ (before the planning horizon begins). The discount factor for the advance term is therefore $\gamma^{s_i - 1 - 1} = \gamma^{-1}$ when the objective is indexed from $t = 1$; equivalently, we index Z from $t = 0$ so the advance contributes $\gamma^0 \cdot A_i$.
- $B_1 = 3 \cdot \sum_i \text{BAC}_i$ (large; cash never binding unless stated).
- $\bar{\eta}_{i,t} = \bar{\eta}$ constant and deterministic unless stated.
- Equal milestone weights $\phi_{i,j} = 1/M_i$ unless stated.
- $\alpha_i = 0$, $\rho_i = 0$ unless stated.
- $\delta^{\text{rec}} = 0.25$, $\psi = 0.10$ (FIDIC defaults) unless stated.
- $\pi_i = 0$ unless stated ($CP_i = \text{BAC}_i$).
- S-curve shape $a_i = b_i = 2$ (symmetric Beta) unless stated.
- Monetary units are abstract.

Note: With $\gamma = 0.95$ and earliest certification dates enforced, the optimal strategy in every completing case is to accumulate progress so that $P_{i,t}^{\text{actual}} \geq \theta_{i,j}$ is first satisfied exactly at $t = e_{i,j}$, then certify immediately when the window opens. Spending earlier than needed wastes discounted cash; spending later than $e_{i,j}$ loses discount value on inflows.

B Group 1: Progress and Milestone Certification

What is being proven: budget converts to progress at rate η, milestones certify at the correct threshold and not before $e_{i,j}$, payments trigger correctly, and milestone ordering is enforced even when progress jumps past multiple thresholds in one period.

B.1 SP-1: Perfect Efficiency, Two Milestones

Economic claim. With $\eta = 1$ and $\gamma = 0.95$, the MILP front-loads spend so that MS1 certifies exactly at $e_{1,1} = 1$. MS2 cannot certify before $e_{1,2} = 3$ regardless of how much progress has been achieved. Deferring the MS2 spend to $t = 3$ saves discounted outflow with identical inflow, so $Z^* = 19.025 > 16.65$ (front-loaded).

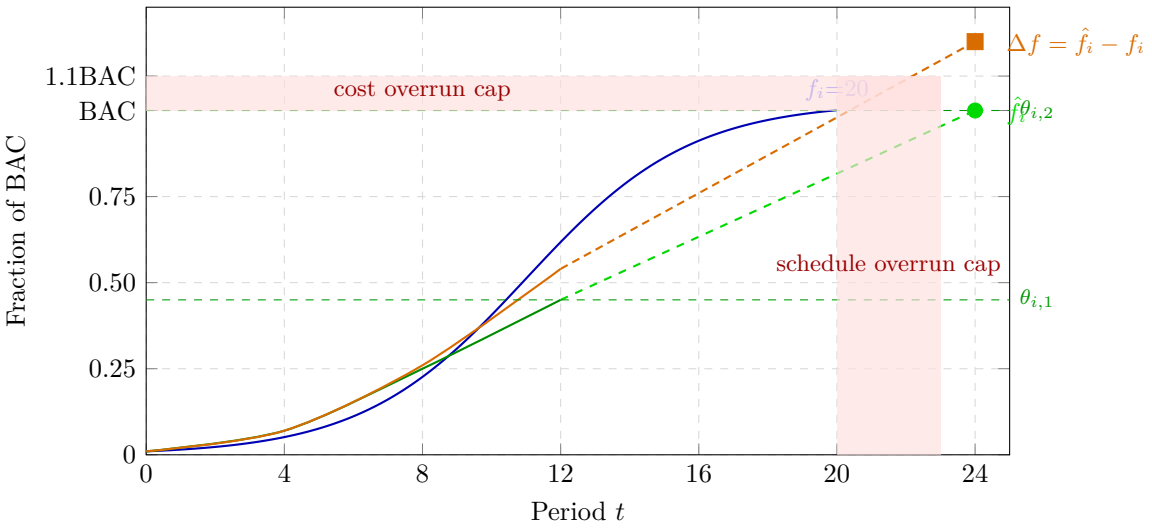


Figure 6: Reference EVM panel. See Section A for legend.

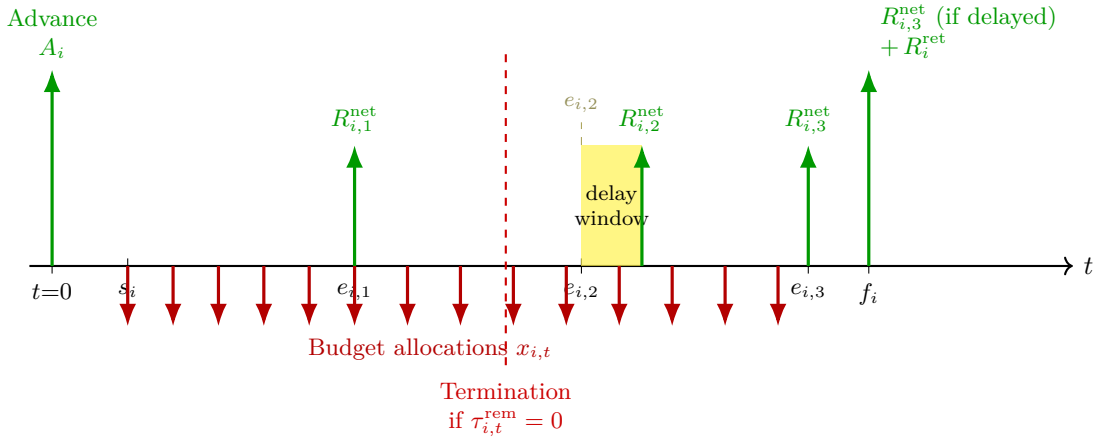


Figure 7: Cash-flow timeline. Legend: Section A.

Parameters.

$n,\ s_1,\ f_1,\ D^{\text{plan}}$	$1,\ 1,\ 4,\ 4$
$\text{BAC}_1,\ CP_1,\ \pi_1$	$100,\ 120,\ 0.20$
$M_1,\ \theta,\ \phi$	$2,\ (0.5, 1.0),\ (0.5, 0.5)$
$\bar{\eta},\ e_{1,1},\ e_{1,2}$	$1.0,\ 1,\ 3$
$\alpha_1,\ \rho_1,\ \Omega_1,\ \mu_1,\ \tau_1^{\text{tol}}$	$0,\ 0,\ 1,\ 0.30,\ 2$
$B_1,\ H,\ \gamma$	$300,\ 4,\ 0.95$

Milestone payments. $R^{\text{net}}_{1,1} = R^{\text{net}}_{1,2} = 60.$

Optimal allocation.

t	$x_{1,t}$	$P^{\text{actual}}_{1,t}$	Window open?	Inflow
1	50	0.50	MS1: ✓	60
2	0	0.50	MS2: × ($e_{1,2} = 3$)	0
3	50	1.00	MS2: ✓	60
4	0	1.00	—	0

$$Z^* = \gamma^0(60 - 50) + \gamma^1(0) + \gamma^2(60 - 50) = 10 + 0 + 0.9025 \times 10 = \mathbf{19.025}.$$

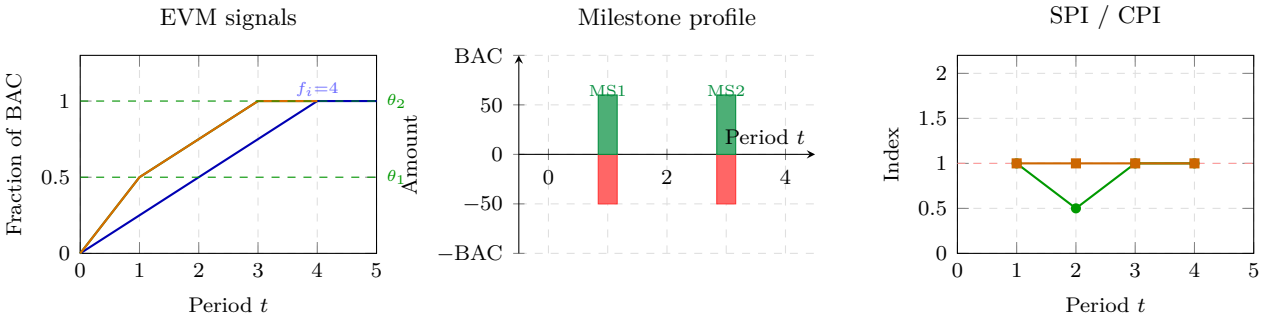


Figure 8: SP-1 project view. Legend: Section A.

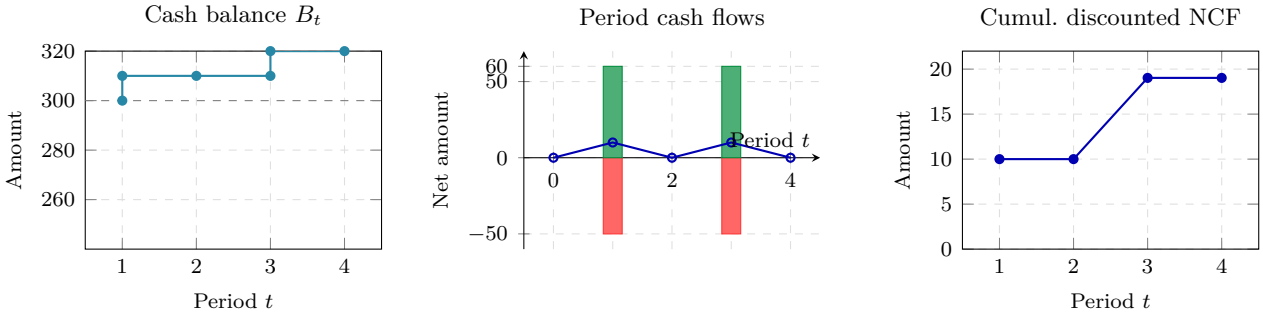


Figure 9: SP-1 portfolio view. $Z^* = 19.025$.

Assertions for pytest.

A1. $x_{1,1} = 50 \pm 0.5$, $x_{1,2} = 0 \pm 0.5$, $x_{1,3} = 50 \pm 0.5$, $x_{1,4} = 0$.

A2. MS1 certified at $t = 1$; MS2 certified at $t = 3$ (not before, since $e_{1,2} = 3$).

A3. $|Z^* - 19.025| \leq 1.0$.

A4. Status: `completed`. No τ decrement.

B.2 SP-3: Three Milestones, $\gamma = 0.95$

Economic claim. With $\gamma < 1$ and earliest certification dates, the MILP certifies each milestone exactly when its window opens, spending just enough just in time to meet the progress threshold at each $e_{i,j}$.

Parameters.

$n,\ s_1,\ f_1,\ D^{\text{plan}}$	$1,\ 1,\ 6,\ 6$
$\text{BAC}_1,\ \pi_1,\ CP_1$	$60,\ 0.50,\ 90$
$M_1,\ \theta,\ \phi$	$3,\ (1/3, 2/3, 1),\ (1/3, 1/3, 1/3)$
$\bar{\eta},\ e_{1,1},\ e_{1,2},\ e_{1,3}$	$1.0,\ 1,\ 3,\ 5$
$\alpha_1,\ \rho_1,\ \Omega_1,\ \mu_1,\ \tau_1^{\text{tol}}$	$0,\ 0,\ 2,\ 0.50,\ 2$

Milestone payments. $R_{1,j}^{\text{net}} = 30$ each.

Optimal allocation. Spend 20 at each $e_{1,j}$.

$$Z^* = \gamma^0 \times 10 + \gamma^2 \times 10 + \gamma^4 \times 10 = 10 + 9.025 + 8.145 = \mathbf{27.17}.$$

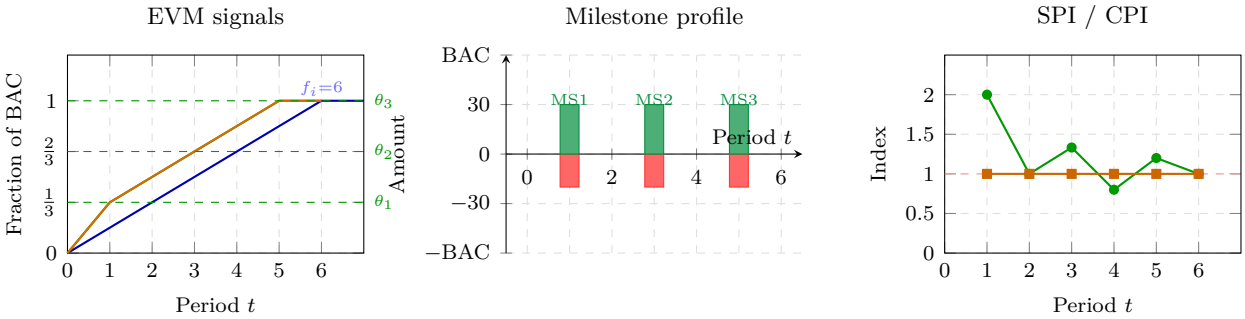


Figure 10: SP-3 project view. Three milestones certified at $t = 1, 3, 5$; spend exactly 20 at each window.

Assertions for pytest.

A1. $x_{1,1} = 20 \pm 0.5$, $x_{1,3} = 20 \pm 0.5$, $x_{1,5} = 20 \pm 0.5$; zero otherwise.

A2. MSj certified at $t \in \{1, 3, 5\}$ respectively.

A3. $|Z^* - 27.17| \leq 1.0$.

A4. Status: `completed`.

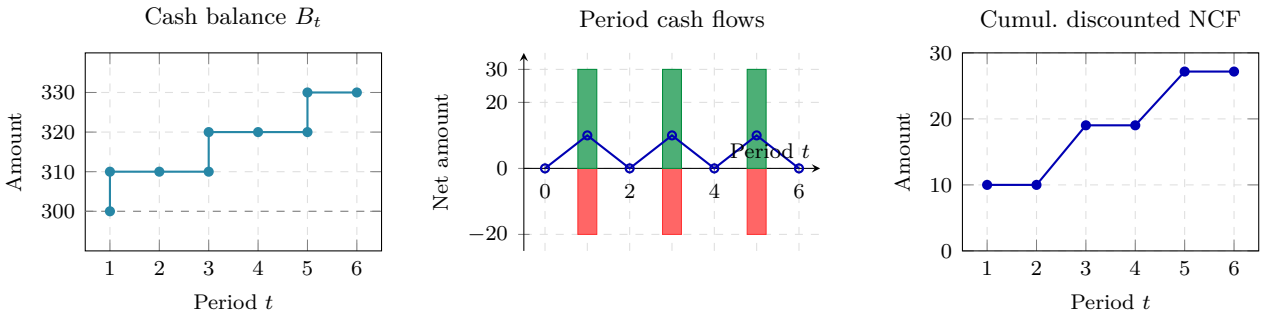


Figure 11: SP-3 portfolio view. Cash balance steps up at each milestone; cumulative discounted NCF reaches 27.17.

B.3 SP-J: Milestone Ordering Enforced on Progress Jump

Economic claim. When a single allocation causes progress to clear both thresholds simultaneously, MS2 still cannot certify before $e_{1,2} = 3$ even though $P_{1,1}^{\text{actual}} = 1.0$ qualifies it. The ordering and earliest-date constraints are independent enforcements.

Parameters.

$n,\ s_1,\ f_1,\ D^{\text{plan}}$	$1,\ 1,\ 4,\ 4$
$\text{BAC}_1,\ \pi_1,\ CP_1$	$100,\ 0,\ 100$
$M_1,\ \theta,\ \phi$	$2,\ (0.5, 1.0),\ (0.5, 0.5)$
$\bar{\eta},\ e_{1,1},\ e_{1,2}$	$1.0,\ 1,\ 3$
$\alpha_1,\ \rho_1,\ \Omega_1,\ \mu_1,\ \tau_1^{\text{tol}}$	$0,\ 0,\ 2,\ 0.50,\ 2$

Optimal strategy. Spend 50 at $t = 1$ (MS1), 50 at $t = 3$ (MS2). All nets zero, so $Z^* = 0$ regardless of discount.

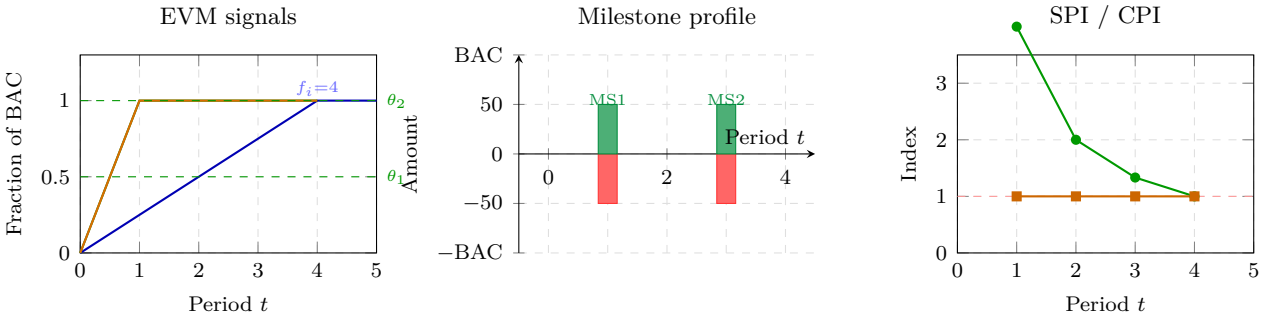


Figure 12: SP-J project view. MS2 certification blocked until $t = 3$ despite progress reaching 1.0 at $t = 1$.

Assertions for pytest.

A1. MS2 not certified at $t = 1$ or $t = 2$ (window $e_{1,2} = 3$ blocks it).

A2. MS2 certified at $t = 3$.

A3. $|Z^* - 0| \leq 1.0$.

A4. Status: `completed` by $t = 3$.

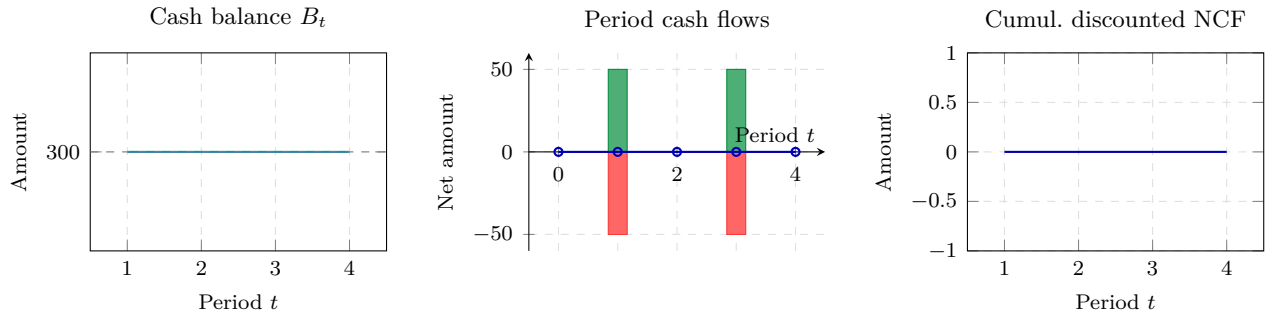


Figure 13: SP-J portfolio view. All nets zero, $Z^* = 0$, cash balance constant at 300.

C Group 2: Discounting and Timing Incentives

What is being proven: $\gamma < 1$ and earliest certification dates together make the optimal allocation uniquely determined. Deferring outflows to the latest period that still enables certification at $e_{i,j}$ is strictly better than front-loading. When a project can be kept alive by partial spend that freezes the termination counter, survival is optimal if and only if the discounted next milestone inflow exceeds the discounted cost of survival allocations.

C.1 SP-4: Deferring Spend to Last Possible Period is Optimal

Economic claim. Deferring the MS2 spend from $t = 2$ to $t = 5$ (when the window opens) saves $\gamma^1 \times 50 - \gamma^4 \times 50 = 6.775$ in discounted outflow with identical inflow timing.

Parameters.

$n,\ s_1,\ f_1,\ D^{\text{plan}}$	$1,\ 1,\ 8,\ 8$
$\text{BAC}_1,\ \pi_1,\ CP_1$	$100,\ 0.40,\ 140$
$M_1,\ \theta,\ \phi$	$2,\ (0.5, 1),\ (0.5, 0.5)$
$\bar{\eta},\ e_{1,1},\ e_{1,2}$	$1.0,\ 1,\ 5$
$R^{\text{net}}_{1,j}$	70 each

$$Z^{\text{opt}} = \gamma^0(70 - 50) + \gamma^4(70 - 50) = 20 + 0.8145 \times 20 = \mathbf{36.29}.$$

$$Z^{\text{front}} = \gamma^0(70 - 50) + \gamma^1(-50) + \gamma^4(70) = 29.515 \quad \text{(strictly dominated)}.$$

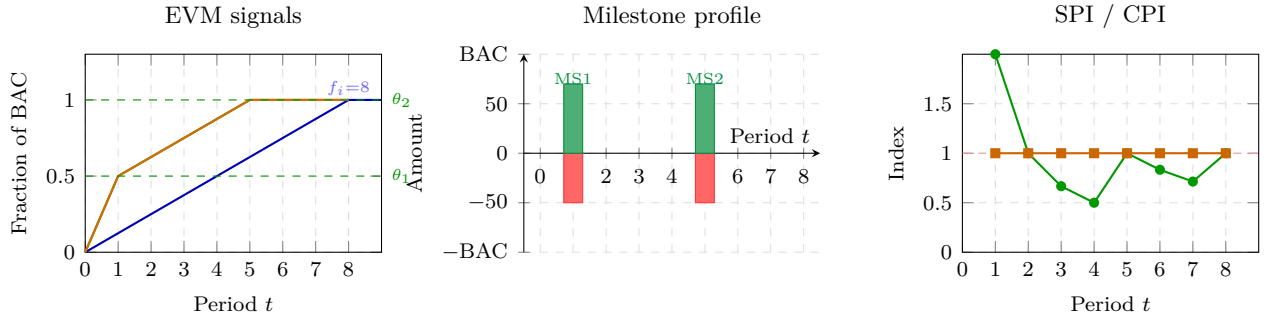


Figure 14: SP-4 project view. MS2 spend deferred to $t = 5$, the exact window opening.

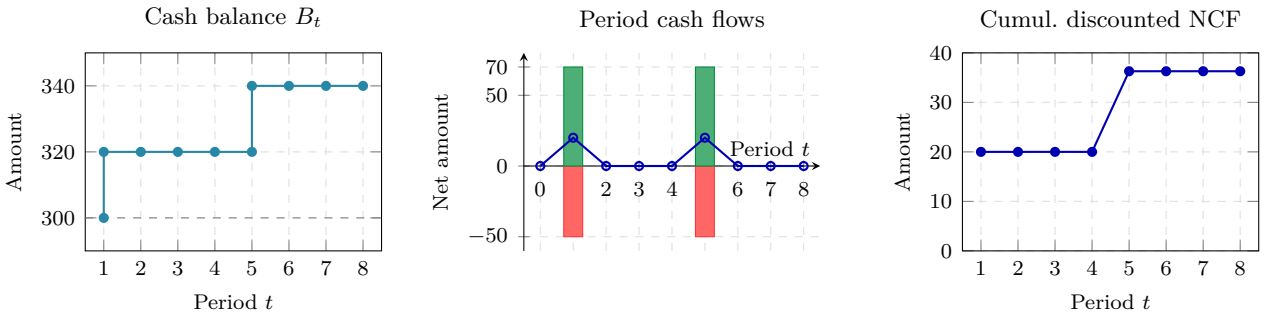


Figure 15: SP-4 portfolio view. $Z^* = 36.29$, larger than front-loaded alternative.

Assertions for pytest.

A1. $x_{1,1} = 50 \pm 0.5$.

A2. $x_{1,2} = x_{1,3} = x_{1,4} = 0 \pm 0.5$.

A3. $x_{1,5} = 50 \pm 0.5$.

A4. MS1 certified at $t = 1$; MS2 certified at $t = 5$.

A5. $Z^* \geq 35.5$.

A6. $Z^* > 29.5$ (front-loaded strategy strictly dominated).

C.2 MP-K: Two Projects, Front-Heavy vs. Back-Heavy Milestones

Economic claim. Project A has a large MS1 payment ($\phi_A = (0.70, 0.30)$) and opens its MS1 window at $e_{A,1} = 1$. Project B has a small MS1 ($\phi_B = (0.30, 0.70)$) opening at $e_{B,1} = 2$. Completing A's MS1 first realises the large early inflow one period earlier, strictly dominating the reverse ordering under $\gamma < 1$.

Parameters.

$n,\ s_i,\ f_i,\ D^{\text{plan}}$	2, 1, 6, 6
$\text{BAC}_i,\ \pi_i,\ CP_i$	60, 0.50, 90
$\phi_A,\ \phi_B$	$(0.70, 0.30),\ (0.30, 0.70)$
$e_{A,1},\ e_{A,2}$	1, 4
$e_{B,1},\ e_{B,2}$	2, 4
$\bar{\eta},\ \Omega_i,\ \mu_i,\ \tau_i^{\text{tol}}$	1.0, 2, 0.50, 2

$$Z_{\text{A-first}} = \gamma^0(63-30) + \gamma^1(27-30) + \gamma^3(90-60) = 33 - 2.85 + 25.72 = \mathbf{55.87}.$$

$$Z_{\text{B-first}} = \gamma^0(27-30) + \gamma^1(63-30) + \gamma^3(90-60) = -3 + 31.35 + 25.72 = 54.07 \quad \text{(dominated)}.$$

Assertions for pytest.

A1. MS1_A certified at $t = 1$; MS1_B certified at $t = 2$.

A2. $Z^* \geq 55.5$.

A3. $Z^* > 54.07$ (B-first strictly dominated).

A4. Both projects `completed`.

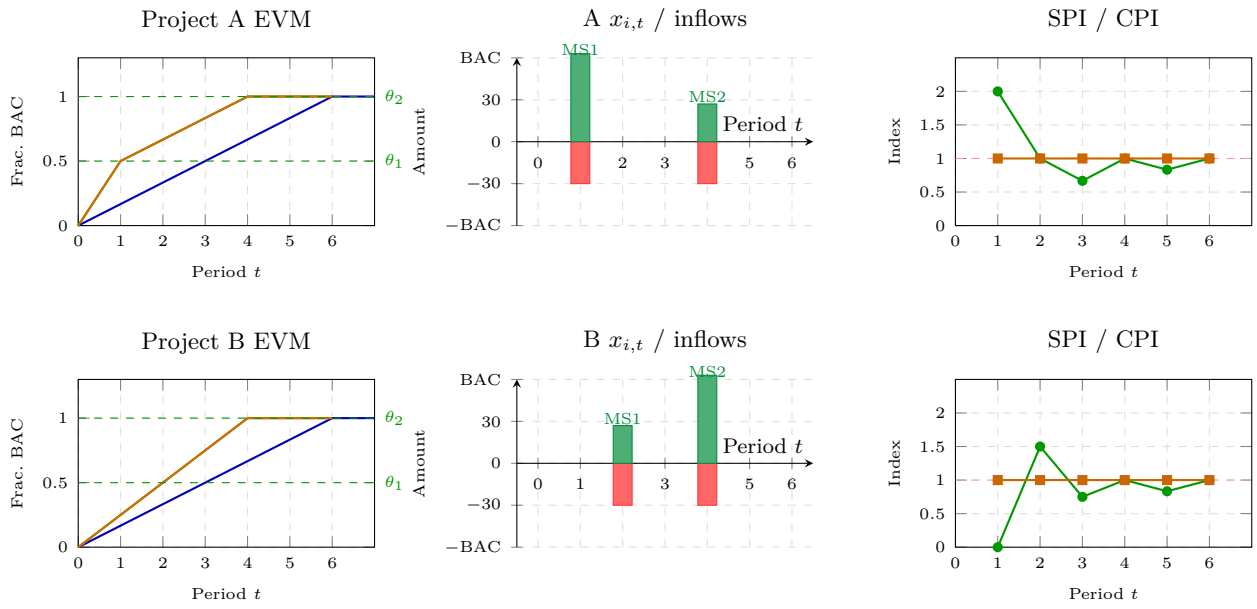


Figure 16: MP-K project view. Project A (top) completes large MS1 first; Project B (bottom) follows.

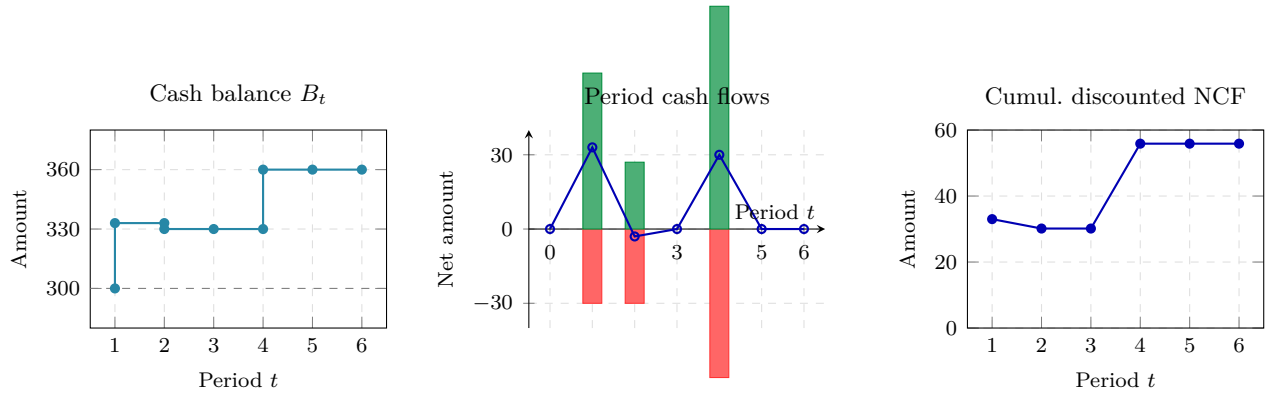


Figure 17: MP-K portfolio view. $Z^* = 55.87$, A-first strictly dominates.

C.3 MP-K′: MP-K with Constrained B_1

Economic claim. With $B_1 = 20$ (sufficient for only one MS1 spend of 15 at $t = 1$), the MILP allocates entirely to A at $t = 1$, starves B until A's MS1 inflow arrives, then funds B's MS1 at $t = 2$. Partial allocation to B at $t = 1$ earns no milestone inflow (progress threshold not met and $e_{B,1} = 2$ not yet open) and wastes discounted cash.

Parameters. Same as MP-K except $B_1 = 20$.

$Z^* = \gamma^0(31.5-15)+\gamma^1(13.5-15)+\gamma^3(13.5+31.5-30) = 16.5-1.425+12.861 = \mathbf{27.936}$.

Note: Spending any amount on B at $t = 1$ (before $e_{B,1} = 2$ opens) accumulates progress but cannot trigger a payment; the discounted outflow is pure waste. The optimal policy is $x_{B,1} = 0$, confirmed by $Z_{\text{partial-B-first}} = 27.686 < 27.936$.

Cash balance trace. $B_1 = 20 \to 20-15+31.5 = 36.5 \to 36.5-15+13.5 = 35.0 \to \cdots \to$ final balance positive throughout.

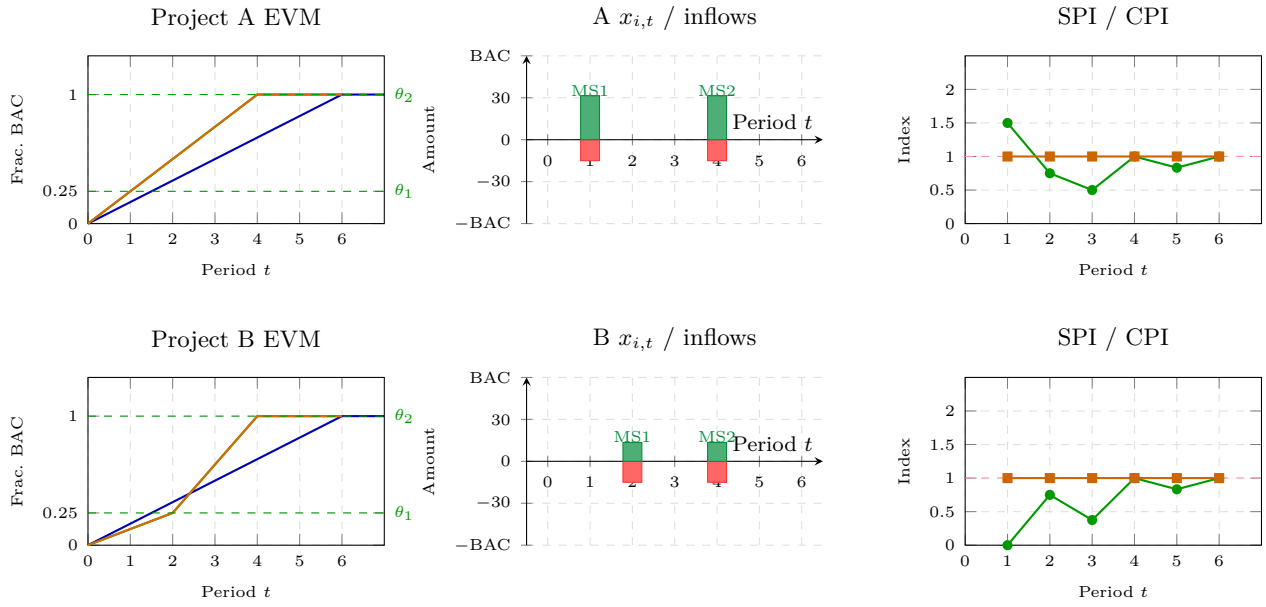


Figure 18: MP-K′ project view. A (top) receives the limited cash first; B (bottom) starved at $t = 1$, funded at $t = 2$.

Assertions for pytest.

A1. $x_{B,1} = 0$ (B starved at $t = 1$).

A2. $B_t \geq 0$ for all t.

A3. MS1_A at $t = 1$; MS1_B at $t = 2$.

A4. $Z^* \geq 27.0$.

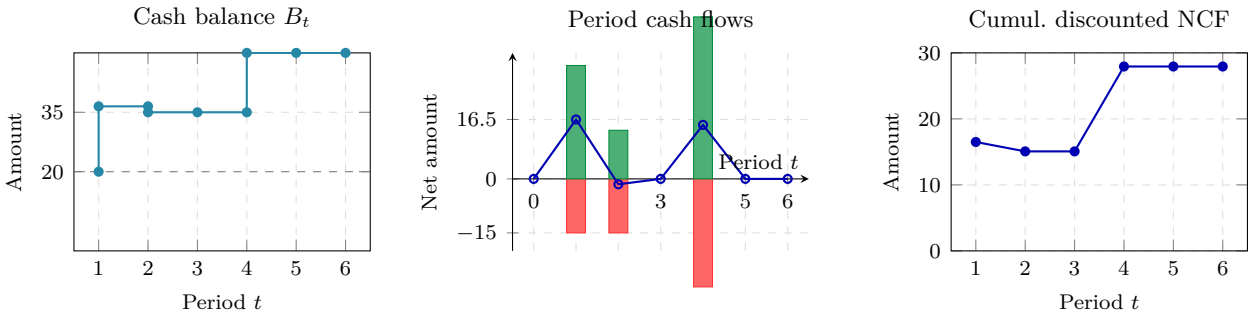


Figure 19: MP-K′ portfolio view. Tight initial cash forces sequential funding; $Z^* = 27.936$.

C.4 SP-L: Survival Spend Optimal (Single Project)

Economic claim. With $\eta = 0.70 < 1$, each unit spent yields only 0.70 units of progress. CPI$= 0.70 < 1$ makes EAC$=$ BAC$/0.70 = 142.9 > (1 + \mu)$BAC, so Condition 2 fires permanently. Spending at rate $\bar{x} = \text{BAC}/(\eta \cdot D^{\text{plan}}) = 17.86$ per period keeps $\text{BCWP}_t = \text{BCWS}_t$ (SPI$= 1$), lapsing Condition 1, so only one condition fires and the counter is frozen. This survival spend is economically optimal: the discounted milestone inflow at $t = 6$ exceeds the total survival cost.

Parameters.

$\text{BAC}_1,\ \pi_1,\ CP_1$	100, 0.70, 170
η_1	0.70 (constant)
$M_1,\ \theta_1,\ \phi_1$	1, 1.0, 1.0
$e_{1,1},\ D^{\text{plan}}$	6, 8
$\mu_1,\ \tau_1^{\text{tol}},\ \Omega_1$	0.10, 2, 1
$\alpha_1 = 0,\ \rho_1 = 0$	

Survival allocation. Spend $\bar{x} = 17.857$ per period for $t = 1, \ldots, 5$ (SPI$= 1$ each period, Condition 1 lapsed, counter frozen). At $t = 6$: spend $x_{1,6} = 100/0.70 - 5 \times 17.857 = 53.571$ to reach $P = 1.0$. MS1 certifies. $R_{1,1}^{\text{net}} = 170$ (no advance, no retention).

$$Z^* = \sum_{t=1}^{5} \gamma^{t-1}(-17.857) + \gamma^5(170 - 53.571) = -68.60 + 77.90 = \mathbf{9.298}.$$

Zero-spend baseline: both conditions fire from $t = 1$; counter $2 \to 1 \to 0$; terminate $t = 3$. $Z = 0$.

Assertions for pytest.

A1. $x_{1,t} = 17.857 \pm 0.5$ for $t = 1, \ldots, 5$.

A2. $\text{SPI}_{1,t} = 1.0 \pm 0.02$ for $t = 1, \ldots, 5$.

A3. $\tau_{1,t}^{\text{rem}} = 2$ for all $t \leq 5$ (counter frozen).

A4. MS1 certified at $t = 6$.

A5. $|Z^* - 9.298| \leq 1.0$.

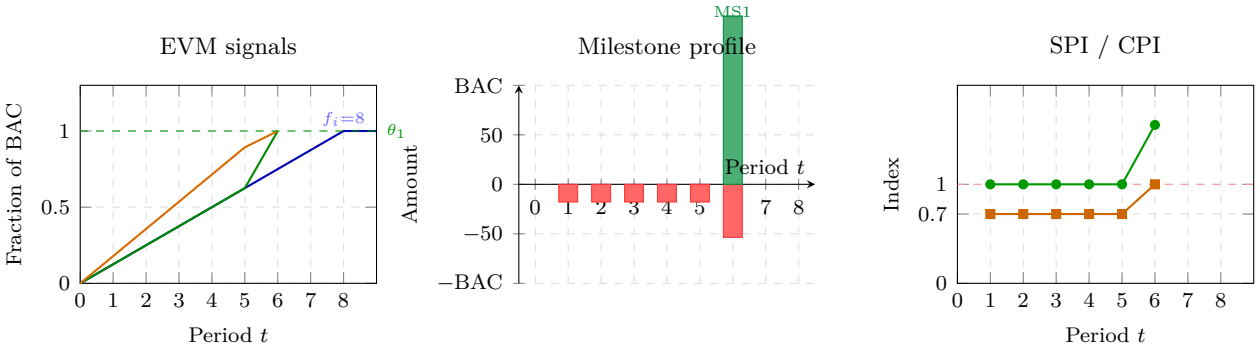


Figure 20: SP-L project view. Survival spend keeps SPI= 1 until $t = 5$; CPI= 0.7 throughout, counter frozen.

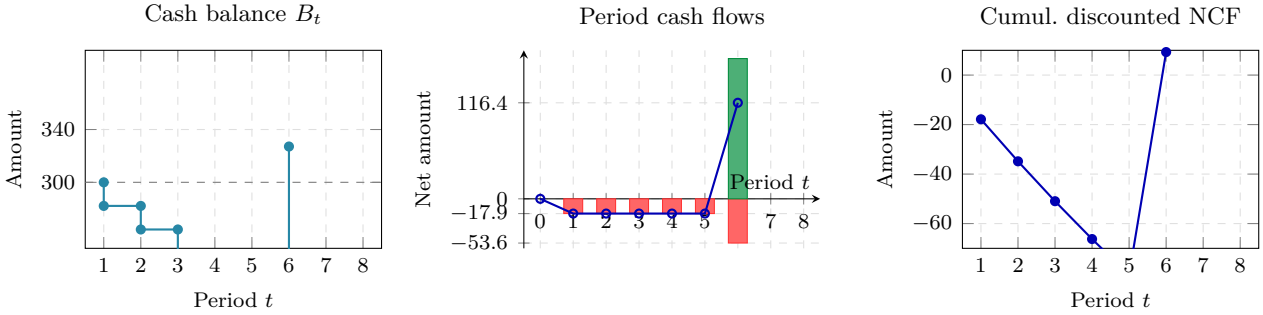


Figure 21: SP-L portfolio view. Survival spend yields positive discounted NCF, $Z^* = 9.298$.

C.5 SP-L': Survival Spend Not Optimal (Single Project)

Economic claim. Same parameters as SP-L except $CP_1 = 80$ ($R^{\text{net}}_{1,1} = 80$). The discounted inflow at $t = 6$ no longer covers the survival cost. Terminating immediately ($Z = 0$) dominates survival.

$$Z_{\text{survive}} = \sum_{t=1}^{5} \gamma^{t-1}(-17.857) + \gamma^5(80 - 53.571) = -68.60 + 18.84 = -49.76 < 0.$$

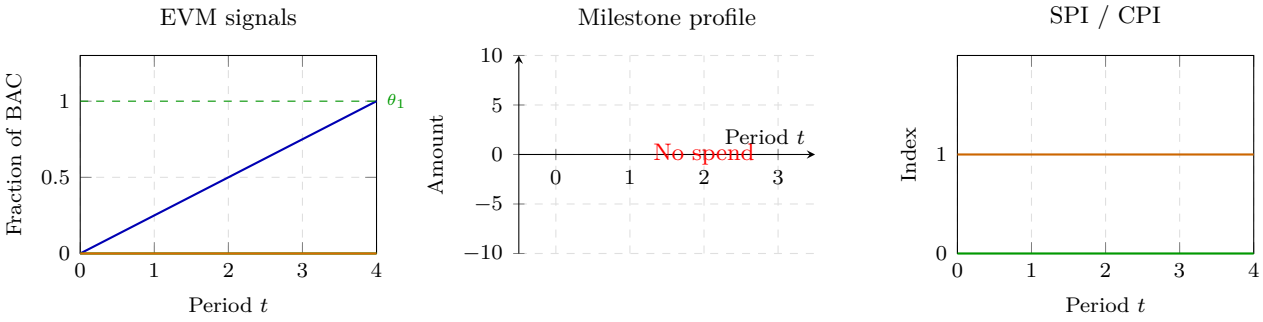


Figure 22: SP-L′ project view. No allocation, immediate termination, $Z = 0$.

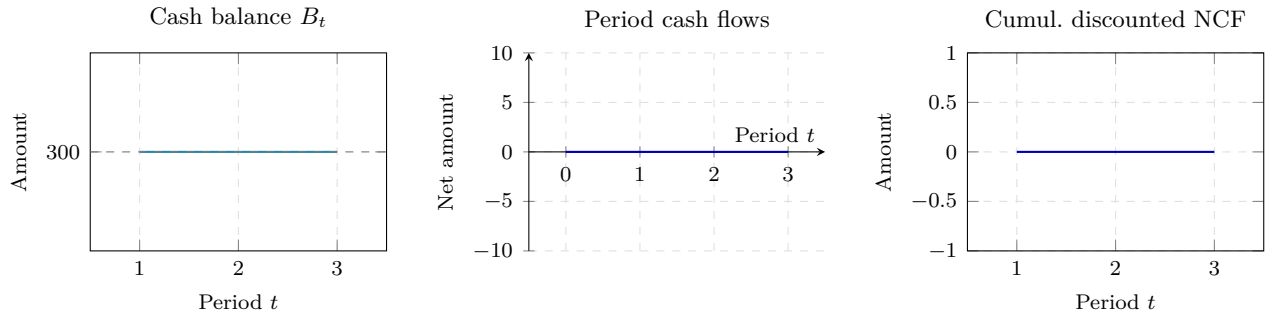


Figure 23: SP-L′ portfolio view. No cash flows, $Z^* = 0$.

Assertions for pytest.

A1. $\sum_t x_{1,t} \leq 1.0$ (MILP allocates essentially nothing).

A2. Project terminates at $t \leq 3$.

A3. $Z^* = 0 \pm 0.5$.

C.6 SP-M: Survival Spend Optimal (Multi-Project)

Economic claim. A Good project (high η, certifies MS1 immediately) generates an early inflow that funds the survival allocations for a Bad project (low η, must survive to $t = 6$ for its milestone). The portfolio value exceeds completing Good alone.

Parameters.

Good: BAC $= 60$, $\eta = 1$, $M = 1$, $\theta = 1.0$, $e = 1$, $CP = 90$	$R^{\text{net}} = 90$, $x_1 = 60$
Bad: same as SP-L above (BAC $= 100$, $\eta = 0.70$, $CP = 170$)	
$B_1 = 90$, $\gamma = 0.95$	

$$Z^* = \underbrace{(90 - 60)}_{Z_{\text{Good}}} + \underbrace{9.298}_{Z_{\text{Bad}}} = 30 + 9.298 = \mathbf{39.298}.$$

Cash balance stays non-negative throughout (verified: $B_t \geq 30$ for all t).

Assertions for pytest.

A1. Good `completed` at $t = 1$.

A2. Bad MS1 certified at $t = 6$.

A3. $B_t \geq 0$ for all t.

A4. $|Z^* - 39.298| \leq 1.5$.

C.7 SP-M': Survival Spend Not Optimal (Multi-Project)

Economic claim. Same as SP-M but Bad's $CP = 80$ (as in SP-L′). The MILP funds only Good and immediately terminates Bad. $Z^* = Z_{\text{Good}} = 30$.

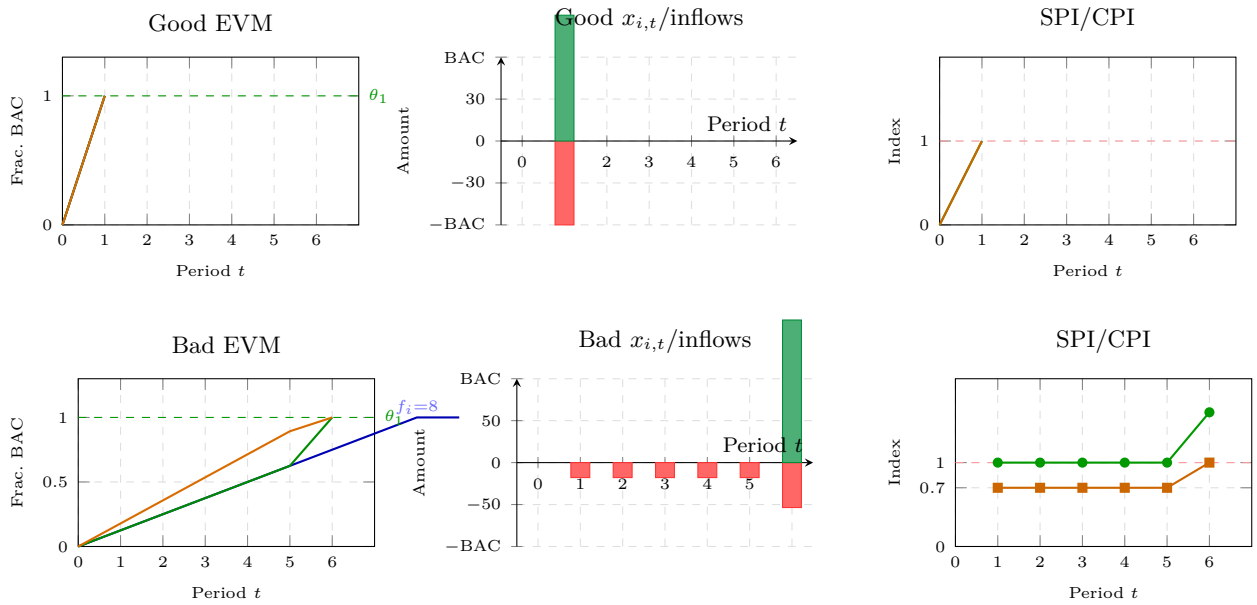


Figure 24: SP-M project view. Good project (top) completes immediately; Bad project (bottom) survives on the released cash.

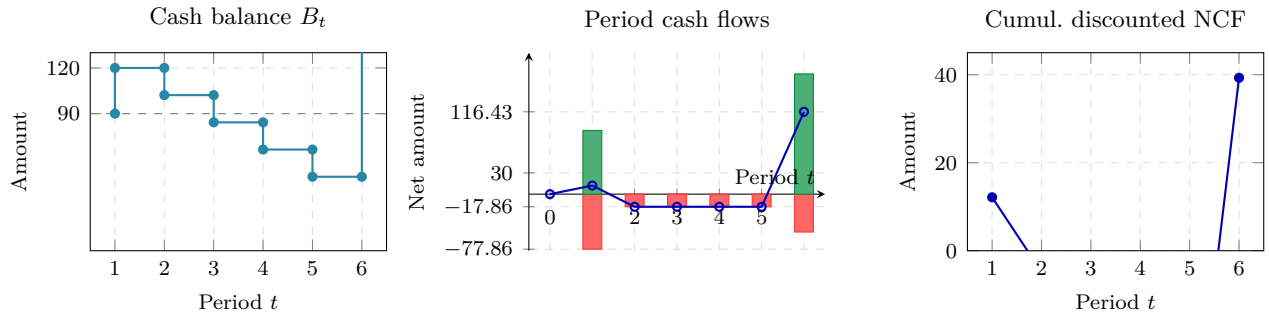


Figure 25: SP-M portfolio view. Good inflow funds Bad survival; $Z^* = 39.30$.

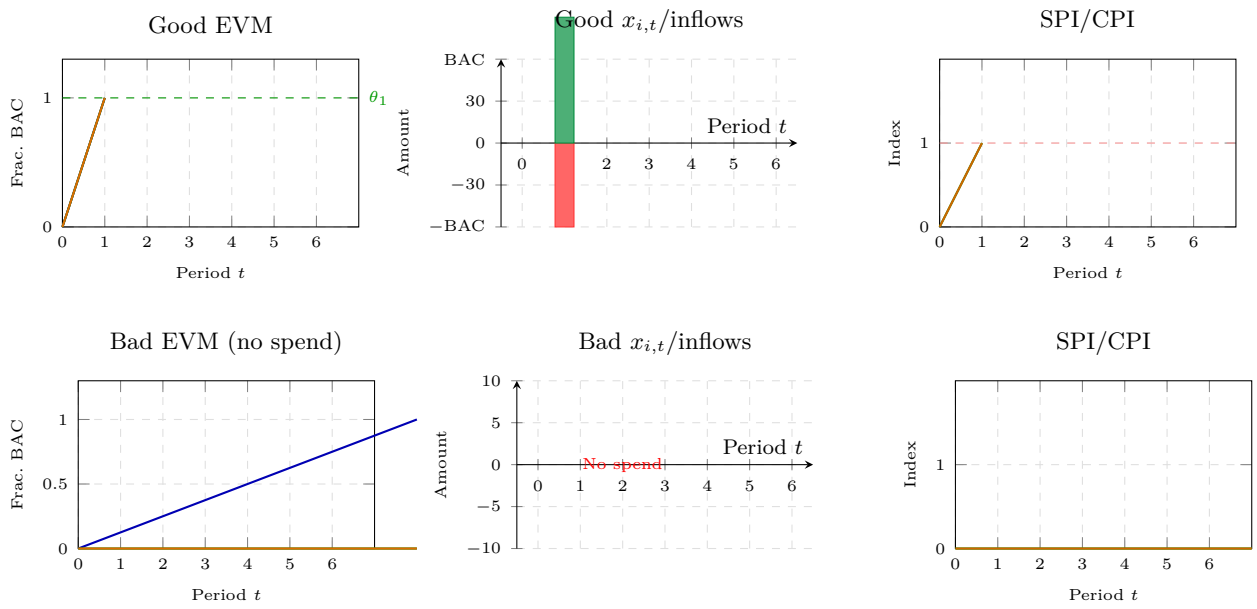


Figure 26: SP-M′ project view. Good completes; Bad starved and terminated.

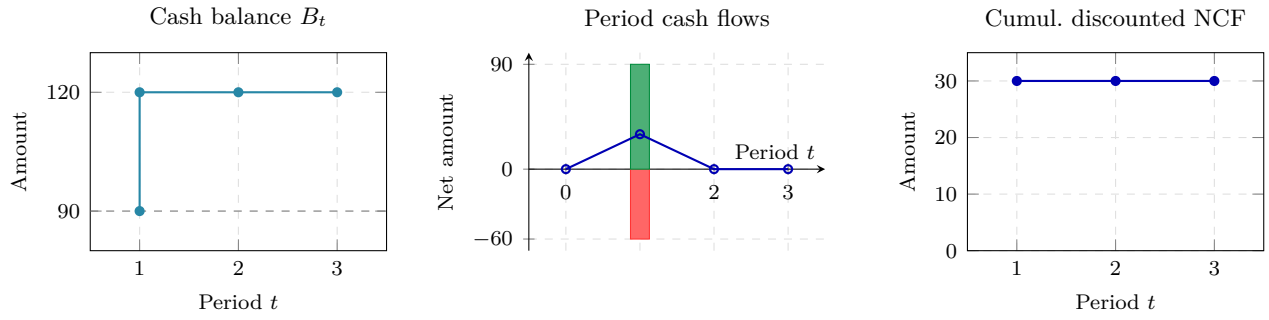


Figure 27: SP-M′ portfolio view. Only Good contributes; $Z^* = 30$.

Assertions for pytest.

A1. Good `completed`.

A2. Bad `terminated` at $t \leq 3$.

A3. $|Z^* - 30.0| \leq 1.0$.

D Group 3: Advance Payment and Recovery Mechanics

What is being proven: advance arrives at $t = s_i - 1$ (period 0 for single-project cases with $s_i = 1$), recovery deductions apply once ψ_i is crossed, deductions stop once advance is fully recovered, and the payment identity $A_i + \sum_j R^{\text{net}}_{i,j} + R^{\text{ret}}_i = CP_i$ holds exactly.

D.1 SP-2: Advance Harvest, Efficiency Decay, Forced Termination

Economic claim. A large advance combined with rapidly decaying efficiency ($\eta_1 = 1.0$, $\eta_{t\geq 2} = 0.05$) makes completion economically impossible. The MILP takes the advance at $t = 0$ (i.e. $s_1 - 1 = 0$) and accepts termination. With $\gamma < 1$, the advance inflow at $t = 0$ is worth more than the discounted settlement at $t = 2$, giving $Z^* = 2.925 > 0$.

Parameters.

$\text{BAC}_1,\ \pi_1,\ CP_1$	100, 0, 100
$\alpha_1,\ A_1$	0.30, 30
$\delta^{\text{rec}},\ \psi,\ \rho_1$	0.25, 0.10, 0
$\Omega_1,\ \mu_1,\ \tau_1^{\text{tol}}$	1, 0.05, 1
$\bar{\eta}$	1.0 at $t = 1$; 0.05 for $t \geq 2$
$H,\ B_1$	6, 300

Zero spend: counter $1 \to 0$ at $t = 1$, termination at $t = 2$. $P = 0$. $R^{\text{term}} = -30$.

$$Z^* = \underbrace{\gamma^0 \cdot 30}_{\text{advance } t=0} + \underbrace{\gamma^2 \cdot (-30)}_{\text{settlement } t=2} = 30 - 27.075 = \mathbf{2.925}.$$

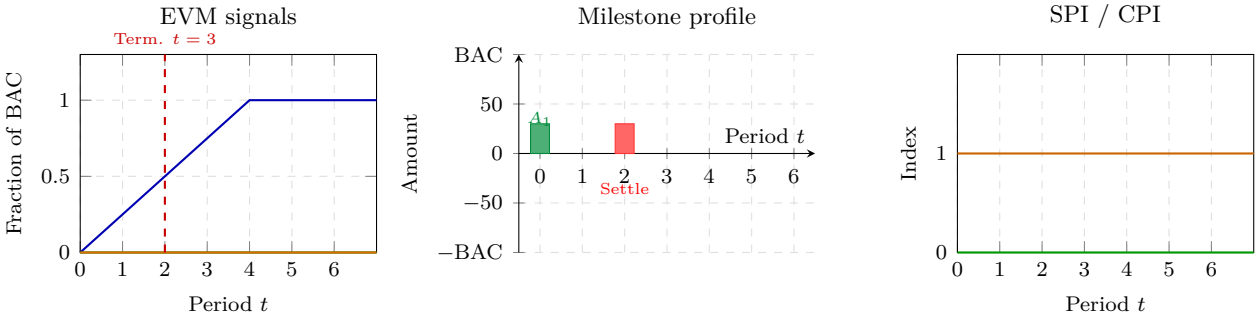


Figure 28: SP-2 project view. Advance taken, zero spend, termination at $t = 3$.

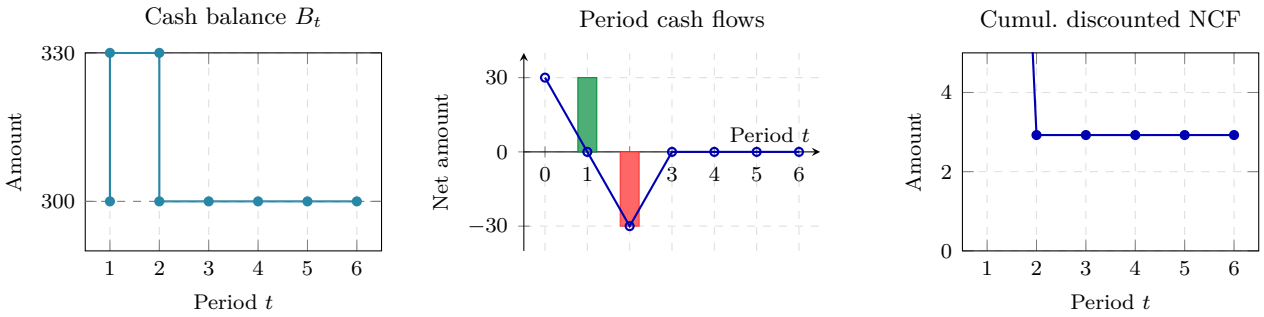


Figure 29: SP-2 portfolio view. $Z^* = 2.925$.

Assertions for pytest.

A1. $\sum_t x_{1,t} \leq 1.0$.

A2. Project terminates at $t \leq 3$.

A3. $R_1^{\text{term}} = -30 \pm 1$.

A4. $|Z^* - 2.925| \leq 0.5$.